MATH3004 Industrial Project

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Final Project Report

Addressing Longevity Risk in Retirement

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Declaration

The work presented in this report is my own work and all references are duly acknowledged.

This work has not been submitted, in whole or in part, in respect of any academic award at Curtin University or elsewhere.

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(Sristi Das)

(27/05/2024)

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Introduction

Retirement is an important phase of one's life in which one's lifestyle is supported by income earned in the past. As the cost of living increases over time due to inflation, it is now more important than ever to have enough accumulated savings at retirement to provide for a retirement lifestyle suited to one's preferred lifestyle. Furthermore, as mortality rates (the rate at which people die each year) lower (e.g. due to medical improvements), the length of time one's accumulated savings need to last is also a critical factor to consider (The Australian Government the Treasury 2019). This report considers critical factors that impact one's retirement, namely longevity/mortality rates, inflation and investment returns, on the expected accumulated savings one would require at retirement age to receive a desired annual income schedule that grows with inflation and the number of years a specified amount of funds are expected to last. The end goal of this research is to provide some rules of thumb as to how increments in these critical factors impact the expected accumulated savings required, and the years until the funds are depleted. For simplicity, retirement age is assumed to be 65 years old for both males and females, and death is assumed to occur at the age of 100, if not before.

Background

To provide some context as to why mortality, inflation and investment returns have been identified as critical factors, the historical variations in each are considered below:

Mortality Risk

To provide some context as to why longevity risk/mortality risk is so important to one's retirement planning, consider Figure 1 below:

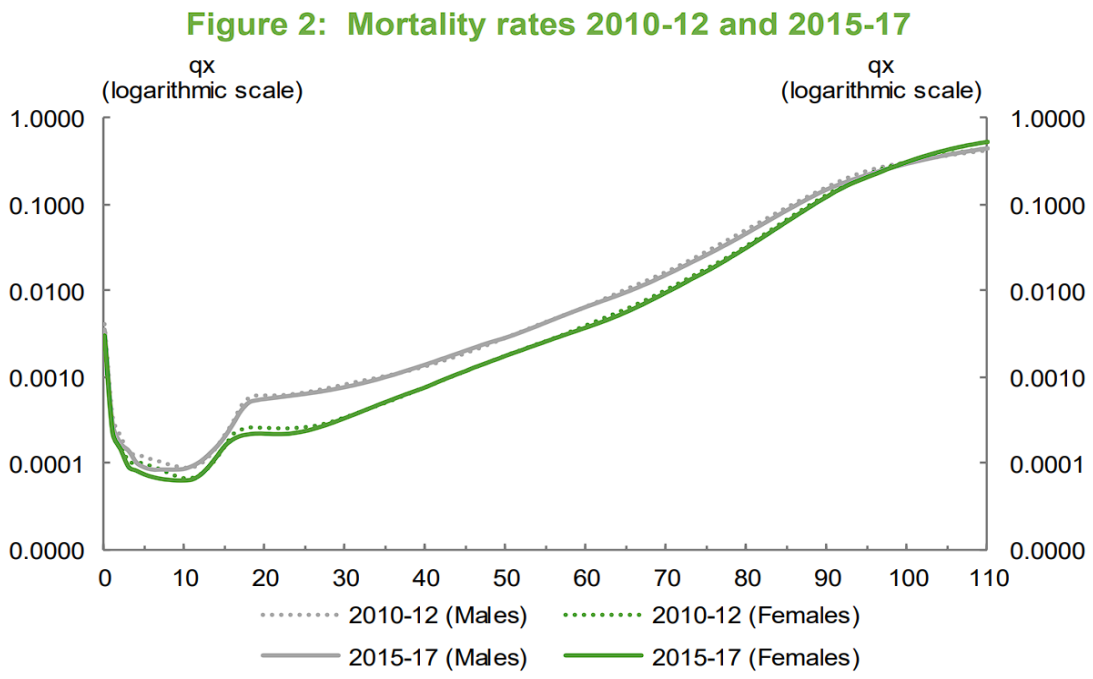

Figure 1: 2011 and 2016 mortality rate comparison. (The Australian Government the Treasury, 2019).

To demonstrate the changes in mortality rates within a few years in the past, Figure 1 above provides the mortality rates for the years 2011 and 2016 respectively for both males and females provided by the Australian Government Actuary (The Australian Government the Treasury 2019). Mortality rate (q_x), representing the rate at which people are dying within one year of their current age, is segregated for both males and females (The Australian Government the Treasury 2019). Female mortality rates are noticeably lower than male mortality rates due to different impacts of illness, lifestyle, and behaviour on either gender, with males more subject to mortality risk due to a greater exposure to these impacts (The Australian Government

the Treasury 2019). The implication of this is that females would require a greater expected accumulated savings amount at retirement age.

Furthermore, the mortality rates for both males and females aged approximately 65 to 100 in 2016 were lower than the corresponding rates for 2013, due to improvements in mortality (The Australian Government the Treasury 2019).

This is further highlighted in Figure 2:

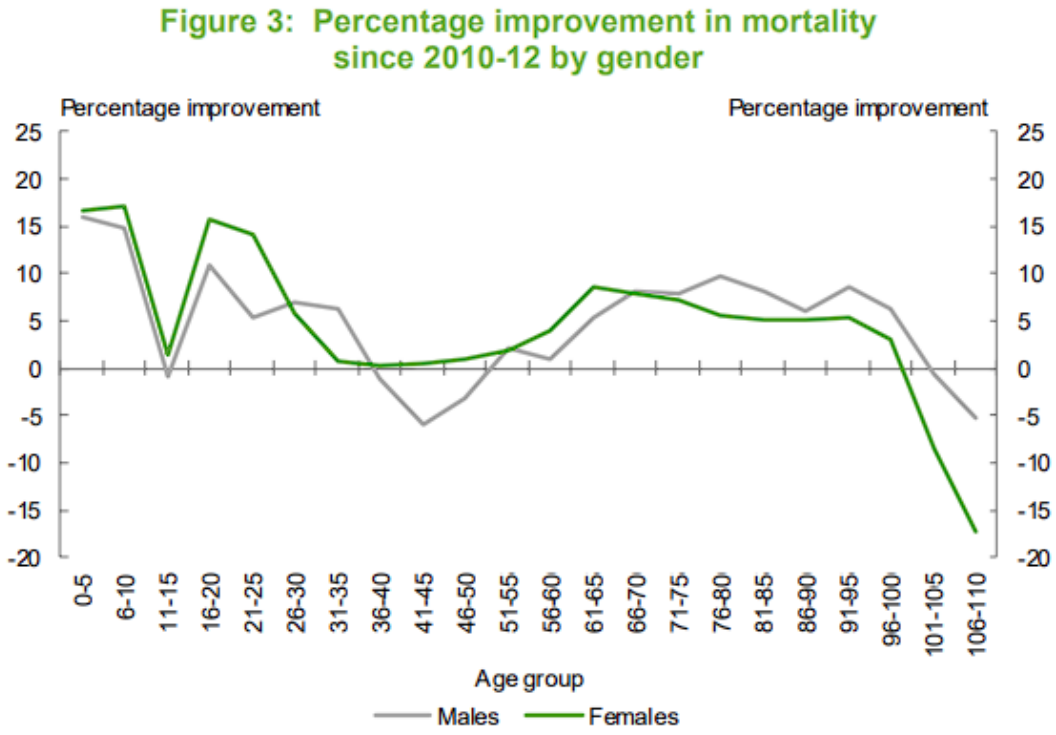

Figure 2: Percentage improvement in mortality for the 3 years between 2013-2016. (The Australian Government the Treasury 2019).

For the ages comprising the retirement phase, ages 65 - 100, there is an improvement in mortality rates for both males and females for all ages within this range. The implication is that if mortality rates continue to improve (reduce in absolute value) at the rates they have in the past, both males and females are exposed to increasing longevity risk, and hence are at a greater risk of outliving their funds.

The effects of mortality improvements are also evident in the historically upward rising trend in life expectancy for males and females aged 65 years old as shown in Figure 3 below, which provides further evidence to suggest that the number of years a person aged 65 (or retirement age) will continue to live on average will steadily continue to increase assuming the historical factors which led to past growth continue to persist in the future (The Australian Government the Treasury 2019).

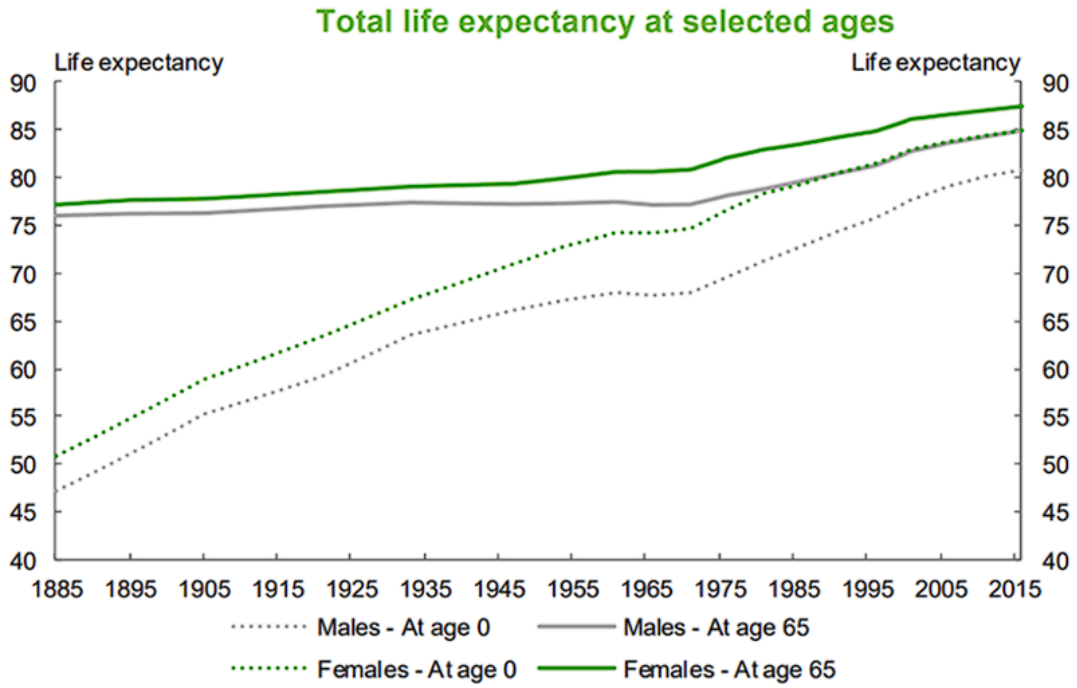

Figure 3: Total Life Expectancy at selected ages (The Australian Government the Treasury 2019).

By factoring in mortality rates, including improvements, in the evaluation of expected accumulated savings, an individual can revise their retirement lifestyle, based on their annual income schedule and circumstances, by accounting for longevity risk.

COVID Effect

Recently, the COVID-19 pandemic has provided a challenge to the stable prediction of the trend in mortality improvements due to the increase in mortality rates due to COVID-19 related

deaths (Actuaries Institute 2023). Based on a comparison of weekly actual deaths by the Mortality Working Group against the weekly predicted deaths over 2020-2023 due to causes excluding COVID-19, and COVID-19 related deaths, Figure 4 shows that predicted weekly deaths have generally been in line or above the weekly deaths with the exception of a few outliers in mostly 2022 (Mortality Working Group 2024I). Figure 5 compares weekly predicted deaths with weekly actual deaths including COVID-19 deaths and COVID-19 related deaths (Mortality Working Group 2024). By comparing Figure 4 with Figure 5, it is evident that the COVID-19 pandemic has played a major role in the number of deaths overall, especially in 2022, where actual deaths far exceeded predicted deaths in Figure 5 compared to Figure 4 with some restabilisation in 2023.

This demonstrates the inherent uncertainty of projecting future mortality improvements into the future, as the impact of COVID in the future as well as future pandemics of a similar nature are not predictable, which makes the evaluation of longevity risk challenging. Future deterioration of mortality rates essentially means that people are more likely to die at any given age compared to some previous year, hence the amount of expected accumulated savings would not need to be as great compared to if mortality rates continued to improve.

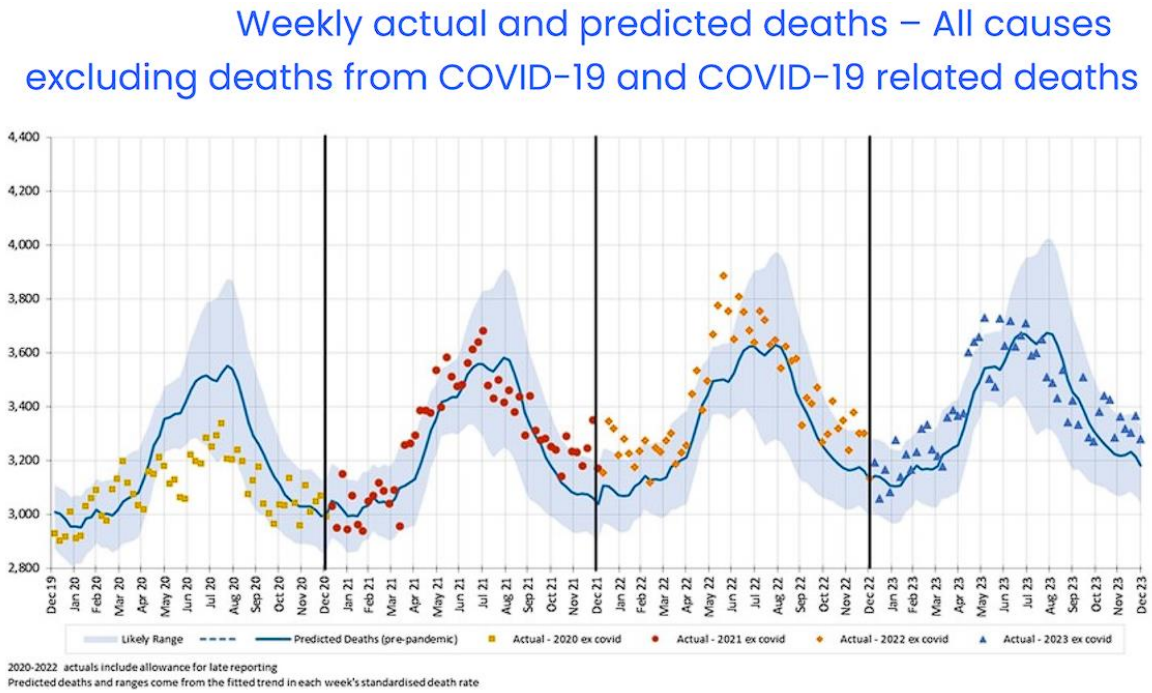

Figure 4: Weekly Actual Death Rate vs Weekly Predicted Death rate for 2020-2023 for all deaths excluding due to COVID-19 and COVID-19 related deaths (Mortality Working Group 2024).

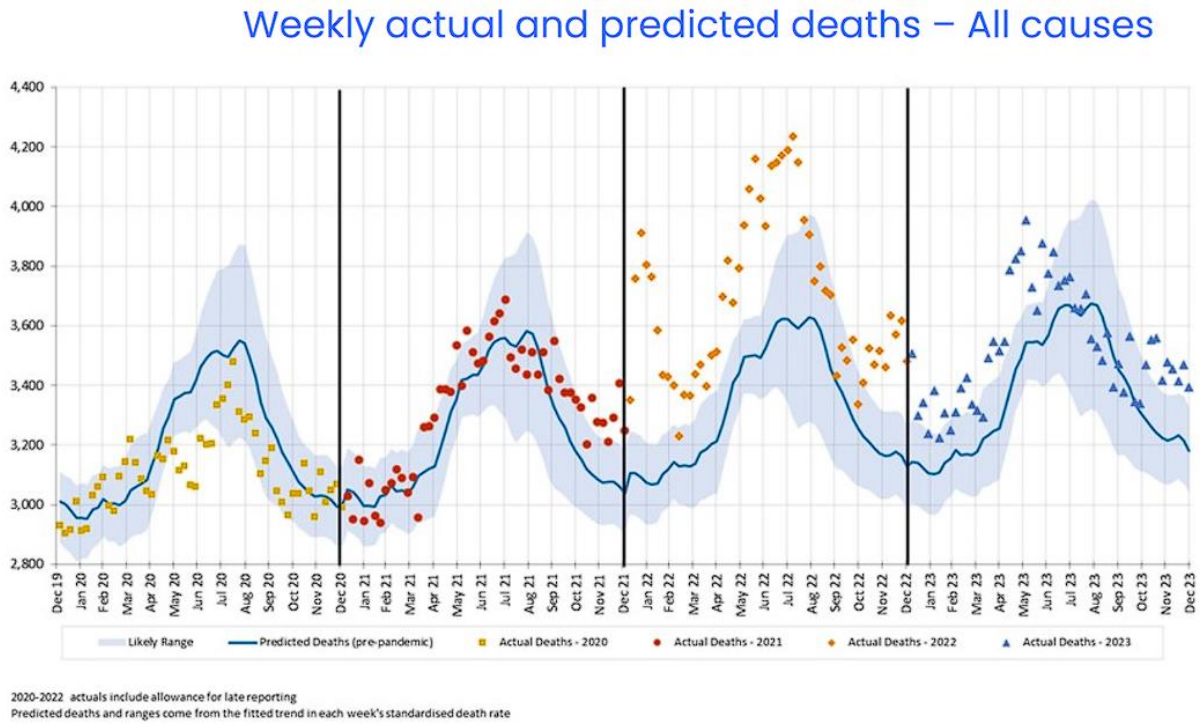

Figure 5: Weekly Actual Death Rate vs Weekly Predicted Death rate for 2020-2023 due to all causes, including due to COVID-19, and COVID-19 related deaths (Mortality Working Group 2024).

Based on analysis conducted by the Mortality Working Group, predictions were made as to how (many deaths would occur for 2023 without the effect of COVID pandemic (Mortality

Working Group 2024). With the results showing that the actual number of deaths are 5% greater than predicted deaths not involving the pandemic effect (Mortality Working Group 2024). Given that COVID-19 has had a deteriorative effect on the expected mortality rate predictions rather than an improvement, a special catastrophic scenario is used for analysis later in this report, where it is assumed that mortality rates deteriorate each year by a certain percentage to compare with other mortality improvement assumptions.

Inflation/Cost of Living Rises

Alongside longevity risk, the cost of living should also be an important consideration when planning for retirement. Assuming that the inflation rate is positive, the cost of living today will be less than the cost of living in the future over the long-term. The Reserve Bank of Australia intends to keep inflation rate between its target range of 2%-3% and historically over the last approximately 30 years, it has had general success in this regard as shown in Figure 6 below (Reserve Bank of Australia, n.d.):

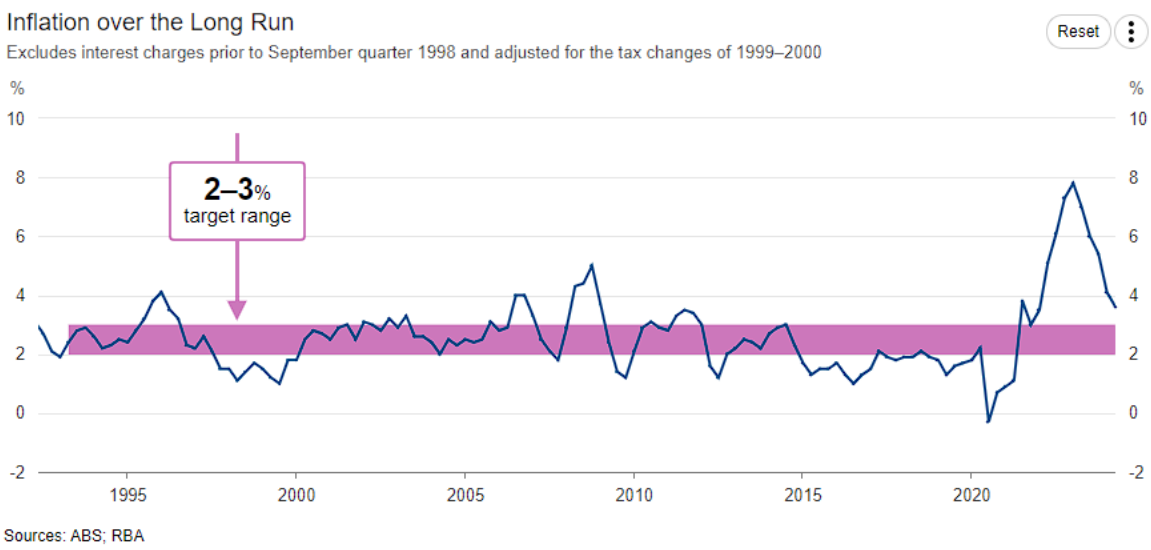

Figure 6: Inflation Rate from June 1992 to March 2024 (Reserve Bank of Australia, n.d).

Hence for the sake of projections, it is assumed that the inflation rate will continue to be within this 2%-3% target range assuming that RBA can manage to keep inflation under control as they have in the past.

Furthermore, the Association of Superannuation Funds Australia (ASFA) has done research regarding a general guideline of the annual expenditure required to live either a comfortable or modest retirement lifestyle (ASFA, n.d.). For an individual aged between 65 - 84 years, the annual incomes required to live a modest and comfortable lifestyle respectively, are $32,665.66 and $51,278.30 in 2023 (ASFA, n.d.). The same individual would have required an annual income of $31,323 and $49,462 respectively in 2022 suggesting an annual rise in the cost of living of approximately 4% for both lifestyles (ASFA 2022).

Investment Returns

The rate of investment return is critical to retirement savings. If the rate of return is too low, an individual's funds may not grow at an appreciable rate to provide for the individual's desired retirement lifestyle. According to ASFA, superannuation funds have provided a 7.1% nominal return over the past 20 years until 30 June 2023 (ASFA 2023).

However, investment returns are subject to wild swings in volatility (ASFA 2023). Although the past 20 years until 30 June 2023 have returned 7.1% nominally by superannuation funds, over the past 5 years the returns were 5.8% nominally, and the past 1 year the return has been 9.2% (ASFA 2023). Assuming the returns stay constant on average over the next 20 years, returns fluctuating from 5.8% to 9.2% represent a fluctuation of 3.4% per annum. To illustrate the significance of this, if $1,000 was invested at the end of each of the next 20 years, the final balance at the end of the investment period would be $36,004 at 5.8%. At 9.2% the final balance would be $52,323, an increase by almost a half. Given that even small percentage changes can have dramatic results on the final value of an investment due to the effect of compounding, fluctuations in investment returns are critical to consider as they could be the difference between having enough funds to last one's retirement or outliving one's funds, and hence requiring other sources of funding to meet the costs of living.

As part returns may not necessarily be indicative of future investment returns, the individual is subject to investment risk in which more expected accumulated savings would be required to fund an individual's lifestyle in order to compensate for the lower investment returns that are possible. Conversely, the greater the investment returns the lower the amount of expected accumulated savings that are required at retirement age.

Age Pension

A potential supplementary source of retirement income, in addition to that derived from superannuation or other savings, is the Age Pension.

An evaluation is conducted based on income and assets to determine the pension that can be paid for an individual receiving the age pension (Department of Social Services 2023). Primarily focusing on the income effect, an individual who is single can earn up to $204 per fortnight, which translates to $5,304 per year, without receiving any pension cuts (Services Australia, 2024). However, once the $204 per fortnight threshold is crossed, the age pension is cut by 50 cents per dollar over the threshold (Services Australia 2024). An individual who is earning above $2,436.60 per fortnight, which translates to $63,352 per year, is cut off from the age pension entirely (Services Australia 2024).

Objectives

The objectives of this research are to:

1. Determine the expected accumulated savings required for retirement given a certain retirement income schedule.
2. Determine the number of years until the funds the individual has saved up for retirement will deplete based on their given retirement income schedule.

3. Develop 'rules of thumb' to estimate the impact of increments in critical factors upon these two measures (that is, expected accumulated savings required and number of years until funds depleted).

The significance of solving these objectives is to help evaluate risks in relation to retirement planning, namely longevity risk, inflation, and investment risk.

Methodology

To solve these two objectives a Discounted Cashflow (DCF) Model was used to generate a cash flow schedule which would determine the expected accumulated savings required for retirement by the following formula:

Expected Present Value (EPV) = $\sum$Retirement Income$_t$ X Probability of Survival$_t$ x Discount Factor$_t$

The expected present value represents the expected accumulated savings that would be required at retirement age while weighting for the probabilities of the likelihood that the cash flow at any given year (t) will be received.

The probability of survival is provided by the following formula:

$$ {}_tp_x = \prod_{k=0}^{t-1} p_{x+k} $$

Where p_{x+k} represents the probability of an individual aged x living until they reach age x + k + 1, and t representing the number of years from age x.

For every p_{x+k}, there is a corresponding mortality rate, denoted by q_{x+k}, that can be calculated by the following formula:

$$ p_{x+k} = 1 - q_{x+k} $$

Where q_{x+k} represents the probability of an individual aged x, dies within 1 year.

In the context of retirement for example, for an individual aged 65 the probability of surviving until their 70th birthday, a 5 year interval, would be denoted as ${}_5p_{65} = {}_1p_{65}$ (or p_{65}) x p_{66} x p_{67} x p_{68} x p_{69}.

Mortality Improvement Formula

$q_x(T) = q_x \text{ X } (1 + I_x/100)^{(T - 2024)}$

The mortality improvement formula allows for the calculation of theoretical mortality rates in the future at year T (YYYY), incorporating future changes in mortality up to that date. For example, if q65 represents the base mortality rate for an individual aged 65 (with the base life table as the 2021 life table), then q65(2024) would represent the hypothetical mortality rate for a 65 year old in 2024. The new mortality rate of q65(2024) is essentially the base mortality rate q65 that has improved (or deteriorated) at a compounded annual rate of I_x since preparation of the base table. For simplicity of the model, it is assumed that the mortality rates have not improved from 2021 to 2024, with mortality rates improving after 2024. As mortality rates start improving after 2024, 2024 is taken as the base year in the mortality improvement formula.

Life Table and Mortality Improvement Sourcing

The mortality rates needed to calculate the Probability of Survival(t) were sourced from the Australian Bureau of Statistics Life Tables, which tabulate various data and statistics regarding mortality rates (Australian Bureau of Statistics 2023). The key life table used was the life table for 2020-2022, which provided the most recent mortality rate data for both males and females from 0 - 100 years of age (Australian Bureau of Statistics 2023). For the purposes of the model,

the data for ages 65 - 100 were used as they were most relevant to the retirement period (Australian Bureau of Statistics 2023).

The mortality improvement rates were sourced from The Australian Government Actuary for both males and females, which provided the most recent past 25 years of mortality rate improvements and the most recent past 125 years of mortality rate improvements (The Australian Government Actuary 2019). As shown in Figure 3, the bulk of the mortality rate improvements for males and females aged 65 have happened in most recent decades when compared to the entire timeline of the past 125 years of mortality rate data until 2015 (The Australian Government the Treasury 2019). For this reason, the past 25 years of mortality rate improvements reflect a greater rate of improvement than the past 125 years of mortality rate improvements (The Australian Government the Treasury 2019).

This data was then used to create new mortality rates for each passing year for an individual, so that the mortality rates used in the calculation of Probability of Survival(t) are always up-to-date in the hypothetical progression of mortality improvements each year. For example, the mortality rates for an individual aged 65 in the year 2020, would have the following mortality rates while factoring for their age each year and mortality improvements:

Age	Mortality Rate
66	q66(2021)
67	q67(2022)
68	q68(2023)
…	…

Life Expectancy Formula

The life expectancy formula is the sum of all ${}_tp_x$ across all the future years that an individual can live. For example, the life expectancy (denoted by e_x) for an individual aged 65 until they die is $e_{65} = {}_1p_{65} + {}_2p_{65} + {}_3p_{65} + \ldots + {}_{35}p_{65}$. This provides a measure as to when an individual aged 65 is likely to die on average (in years), which will be useful when comparing with how soon they will run out of funds. If their funds run before their life expectancy, the individual may want to prepare for this possibility by considering other sources of funding or by reducing annual retirement income (that is, moderate lifestyle expectation). However, the life expectancy is an average, any one individual may live longer than the life expectancy or shorter than the life expectancy, and the life expectancy is done across the whole cohort of individuals and does not account for specific lifestyle choices and illnesses that any one individual may have. Nevertheless, the life expectancy calculation still provides a useful benchmark to measure how soon an average person will run out of funds relative to when the average individual will die.

Discount Rate

The discount rate is set equal to the assumed investment rate of return. This allows future cash flows to be expressed in today's dollar values, allowing for comparison of investment options. For example, assuming that 10% is the desired rate of return, $1.10 in one year's time is equivalent to $1 in the hands of the individual today. By discounting the future expected cash flows that an individual will receive over their retirement phase to today's dollars, we are able to determine how much funds an individual will require at retirement age assuming a specified rate of return.

Years to depletion of fund

To calculate the number of years the funds will deplete, the individual's lump sum amount at retirement age were compared to a running total of the EPV of retirement income for each additional year (rounded down to the nearest year).

For example, consider a specified retirement income of $100,000 per annum and a lump sum of $175,000 at retirement. Then, assuming an investment return of 10% per annum with no growth in cash flows or mortality improvement:

Year (t)	Nominal Cash Flow (t)	Present Value of Cash Flow (t)	Running Total of PV (t)
1	$100,000	$90,909	$90,909
2	$100,000	$82,645	$173,555
3	$100,000	$75,133	$248,688

The expected accumulated savings required to fund this annual income schedule at the beginning of the investment period would be $248,688 for the specified conditions. Given that the individual has only $175,000 at the start of the period, they are underfunded for their desired lifestyle. If they were to continue with this lifestyle, the number of years it would take for the funds to deplete would be closest to two years (rounded down) as $90,909 + $82,645 = $173,555 which is less than $175,000, drawing down from the interest they receive from their investments and the principal.

DCF Model

A DCF Model created in Microsoft Excel was used to model the cash flows for an individual based on input values regarding the variables for the retirement income schedule, and variables regarding the individual themselves along with their associated mortality rate factors.

The variables regarding the individual were:

1. Gender: Male/Female, and
2. Age: 0 - 100

The variables regarding the cash flow schedule were:

1. Starting Annual Income
2. Discount Rate
3. Inflation Rate

As the annual income increases each year at the rate of inflation, the growth rate and the inflation rate are equal.

The variables regarding the mortality rate were:

1. Formula Year - The year in which mortality improvement rates are used, excluding for the case of a catastrophic scenario (worst case).
2. Adjusting Year - The year the mortality rates are improved to.
3. Catastrophic Scenario - An arbitrary rate at which mortality rates are assumed to have deteriorated at 3% p.a. 5% was intended to be used however the mortality rates generated by using 5% lead to rates which are greater than 1, which are not possible as the number of deaths in a cohort can not exceed the number of individuals in the cohort. 3% is found to be a mortality deterioration rate that is close to the upper limits of the mortality deterioration rates that can be used for analysis without generating a model where mortality rates exceed 1 for any given age above 65 years old for either male or female.

4. The Mortality Improvement Type - assuming No Mortality Improvement, Past 25 Years of Improvement, Past 125 Years of Improvement, or the Catastrophic Scenario of Improvement (based on the arbitrary rate discussed above in variable 3.).

Scenario testing was conducted to investigate the effects of incremental changes in assumptions in mortality improvement type, discount rate, and inflation rate within a certain range of increments. 48 scenarios were created for various combinations of assumptions of the mortality improvement type, the discount rate, and the inflation rate as shown below for sensitivity testing:

Scenario	Mortality Improvement Type	Discount	Inflation Rate
1	No Mortality Improvement	6.00%	2.00%
2	No Mortality Improvement	6.00%	2.50%
3	No Mortality Improvement	6.00%	3.00%
4	No Mortality Improvement	7.00%	2.00%
5	No Mortality Improvement	7.00%	2.50%
6	No Mortality Improvement	7.00%	3.00%
7	No Mortality Improvement	8.00%	2.00%
8	No Mortality Improvement	8.00%	2.50%
9	No Mortality Improvement	8.00%	3.00%
10	No Mortality Improvement	9.00%	2.00%
11	No Mortality Improvement	9.00%	2.50%
12	No Mortality Improvement	9.00%	3.00%
13	Past 25 Years	6.00%	2.00%
14	Past 25 Years	6.00%	2.50%
15	Past 25 Years	6.00%	3.00%
16	Past 25 Years	7.00%	2.00%
17	Past 25 Years	7.00%	2.50%
18	Past 25 Years	7.00%	3.00%
19	Past 25 Years	8.00%	2.00%
20	Past 25 Years	8.00%	2.50%
21	Past 25 Years	8.00%	3.00%
22	Past 25 Years	9.00%	2.00%
23	Past 25 Years	9.00%	2.50%
24	Past 25 Years	9.00%	3.00%
25	Past 125 Years	6.00%	2.00%
26	Past 125 Years	6.00%	2.50%
27	Past 125 Years	6.00%	3.00%
28	Past 125 Years	7.00%	2.00%
29	Past 125 Years	7.00%	2.50%
30	Past 125 Years	7.00%	3.00%
31	Past 125 Years	8.00%	2.00%
32	Past 125 Years	8.00%	2.50%
33	Past 125 Years	8.00%	3.00%
34	Past 125 Years	9.00%	2.00%
35	Past 125 Years	9.00%	2.50%
36	Past 125 Years	9.00%	3.00%
37	Catastrophic	6.00%	2.00%
38	Catastrophic	6.00%	2.50%
39	Catastrophic	6.00%	3.00%
40	Catastrophic	7.00%	2.00%
41	Catastrophic	7.00%	2.50%
42	Catastrophic	7.00%	3.00%
43	Catastrophic	8.00%	2.00%
44	Catastrophic	8.00%	2.50%
45	Catastrophic	8.00%	3.00%
46	Catastrophic	9.00%	2.00%
47	Catastrophic	9.00%	2.50%
48	Catastrophic	9.00%	3.00%

Figure 7: Combinations of scenarios for scenario testing and sensitivity analysis

The EPV was generated for each scenario and the number of years until the funds were depleted were calculated alongside the EPV.

Results

Data Preparation

The external data required for the DCF Model calculations were primarily, the 2020 - 2022 ABS Life Tables for both males and females and the 2015-2017 mortality improvement rates. No data preparation was required to analyse the data, as the raw data were direct inputs to generate the simulated data for the EPV calculations and Funds Depletion calculations for the various scenarios.

Discussion

To make analysis of various changes in assumptions easier, the following standardised individual is used for Objective 1: A male/female aged 65 years old will get $100,000 in their starting annual income with their income increasing each year at the rate of inflation. The reason for using $100,000 for the starting annual income is because it provides a base to calculate the expected annual savings required at retirement age more easily intuitively. For example, if an individual instead wants $50,000 as their starting annual income, the expected accumulated savings amount for a $100,000 starting annual income as shown in the results of the model can be halved. If the individual wants $200,000 as their starting annual income, the expected annual savings amount can be doubled.

Figure 8 below shows a general overview of the accumulated savings required for a male at retirement age for a $100,000 starting annual income that grows with inflation for various assumptions of mortality improvement, inflation rate, and discount rate as described in the scenario table above (Figure 7).

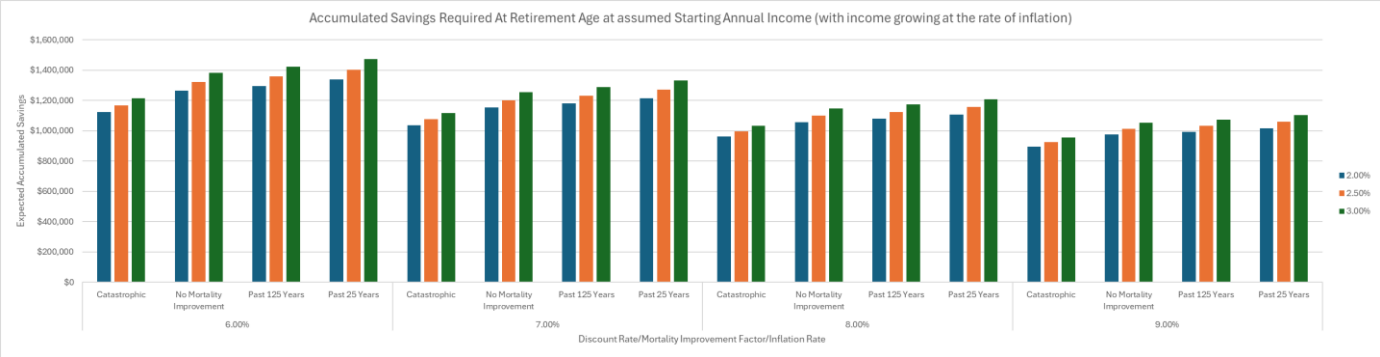

Figure 8: Expected Accumulated Savings Required for various scenarios

For given discount and inflation rates, the EPV required for catastrophic is lowest compared to no mortality improvement, the past 125 Years of Mortality improvement and the past 25 years of mortality improvement, as the individual would on average die sooner, thus requiring less funds initially.

For increases in mortality improvement type, the EPV increases as more funds will be required as the individual is more likely to live longer and hence will require more funds to sustain their longevity. This is seen on average for a select discount rate, say 6%, where as the mortality rates improve from catastrophic to the past 25 years, the amount of funding required for retirement increases in an upward trend for each inflation rate.

For given mortality improvement and inflation rates, for every increase in discount rate, the funds required at retirement decreases, as a greater investment rate of return compensates for having a lower amount of funds at retirement age. This is seen across corresponding mortality improvement types and inflation rate scenarios. In other words, for a catastrophic mortality improvement scenario and 2% inflation, the amount of accumulated savings required at retirement age decreases as the discount rate increases.

For given mortality improvement and discount rates, increases in inflation rate will mean that more funds are required to accommodate the increasing cost of living. For any given discount

rate and mortality improvement type, say 6% and Catastrophic mortality improvement, the amount of funds increases as the inflation rate increases.

A similar graph was prepared for a female aged 65. However, the relative difference resulting from a change to the 3 critical factors is similar to that for males. Hence the graph for females has been left to the appendix.

The following graphs help in answering Objective 3 in determining the effects of incremental increases in retirement factors on the effects on the expected accumulated savings that would be required at retirement age. For example, Figure 9 shows that an incremental increase of 0.5% in inflation rate would result in a 3.14% increase in the amount of funds required at retirement age on average, assuming No Mortality Improvement, and a 6% discount rate. This incremental effect calculation is done for a specific scenario where mortality improvement type and discount rate are fixed, with the variable being investigated being the inflation rate. Discussions of the effect of incremental increases in one changing variable (with others remaining constant) on expected accumulated savings across all scenarios on average is provided below.

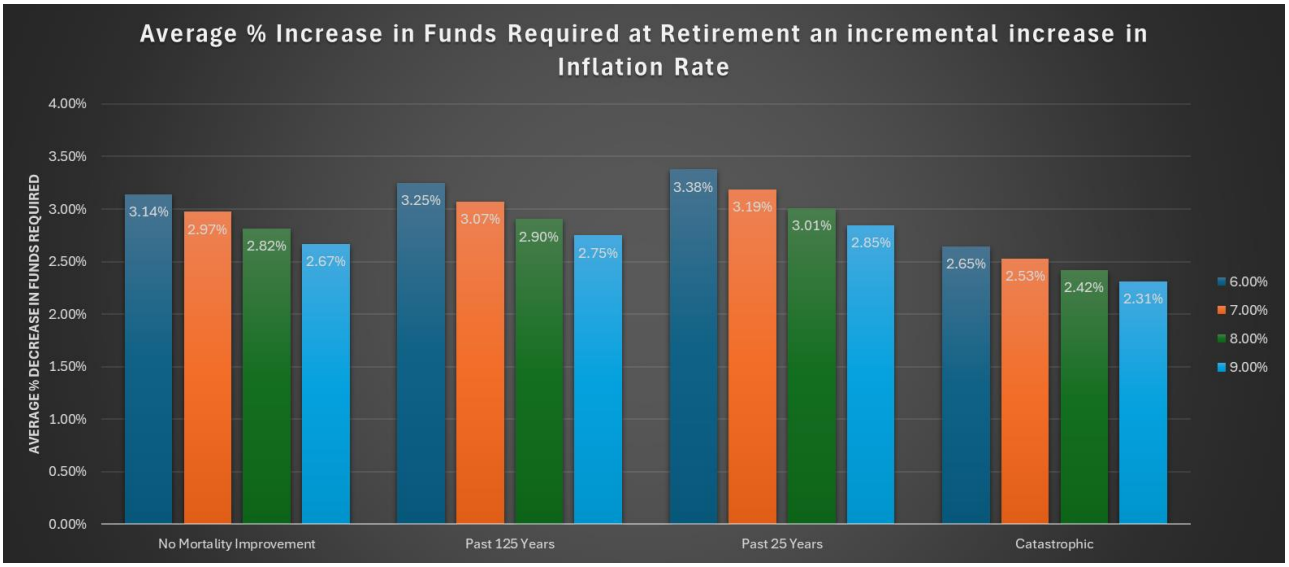

Figure 9: Average % Increase in Funds Required at Retirement for specified conditions and desired annual income schedule for an incremental increase in Inflation Rate for males

The graph above shows that for increments in the inflation rate of 0.5%, on average increase the funds required for retirement at 65 by about 2.9% for varying mortality improvements types and discount rates for males. For females this is 3.1%. The graph for females is similar to the graph for the males with slightly different values and has hence been put in the appendix.

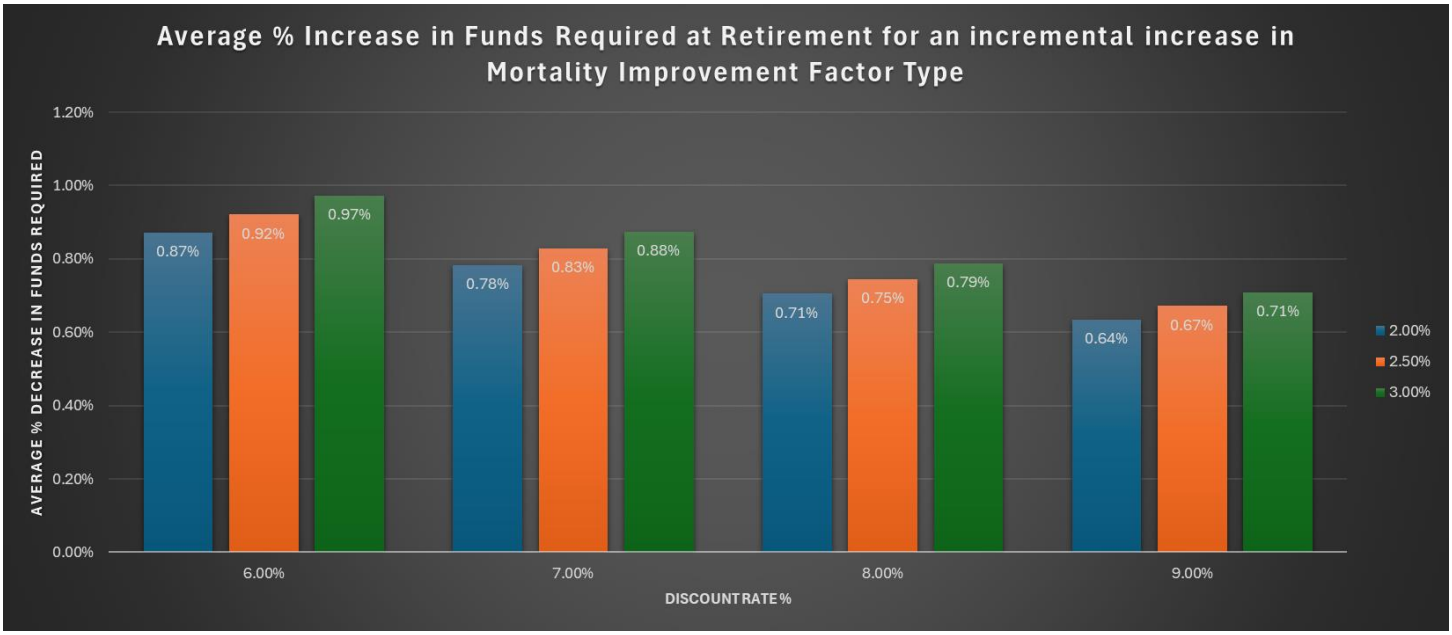

Figure 10: Average % Increase in Funds Required at Retirement for specified conditions and desired annual income schedule for an incremental increase in Mortality Improvement Factor Type for males

The graph above shows that for increments in mortality improvement type, on average increase the funds required for retirement at 65 by about 1.4% for varying mortality improvements types and discount rates. For females this is 1.2%. The graph for females is similar to the graph for the males with slightly different values and has hence been put in the appendix.

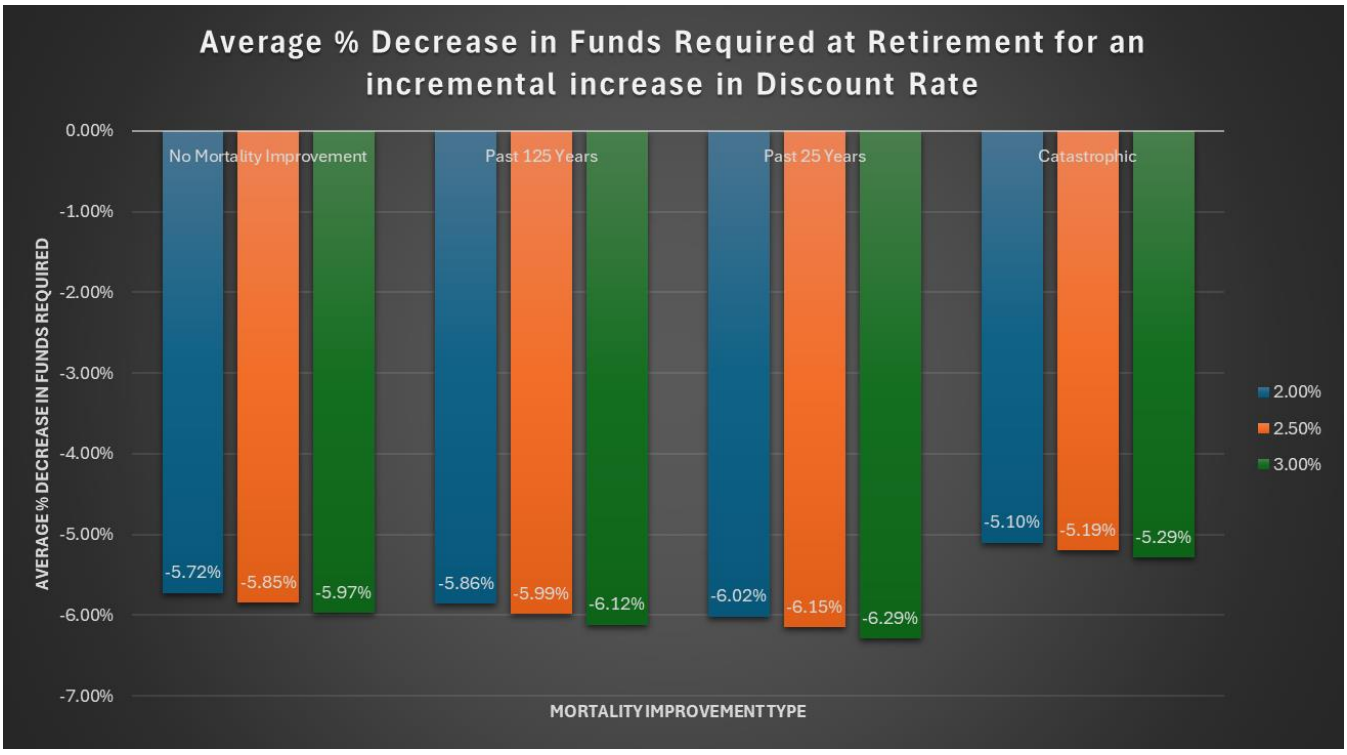

Figure 11: Average % Decrease in Funds Required at Retirement for specified conditions and desired annual income schedule for an incremental increase in Discount Rate for males

The graph above shows that for 1% increments in discount rate, on average decrease the funds required for retirement at 65 by about 5.8% for varying mortality improvements types and inflation rates. For females this is about a 6.1% decrease. The graph for females is similar to the graph for the males with slightly different values and has hence been put in the appendix.

Years Until Funds are Depleted

For Objective 2, the standardised individual is a male/female aged 65 years old with $1,000,000 in lump sum at the beginning of retirement with an annual starting expenditure of $100,000 growing with inflation.

The green line represents the life expectancy of a male 65 years old. This provides a benchmark against which to test whether an individual will outlive their funds by a comparison of their life expectancy against years to deplete the funds for the various scenario testing cases. The life expectancy line can be considered an measure of how long the individual can be expected to live. While the years until the funds deplete can be considered the "lifespan" of the funds based on the annual expenditure. If life expectancy > years to deplete, then the individual will outlive their funds on average. Conversely, if years to depletion > life expectancy, the individual's funds will not deplete before their lifespan on average, and hence they will have plenty of funds in reserve.

In certain parts of the graph, the years until the funds deplete at 35 years, which is the maximum number of years a 65 year old can live in this model (as it is assumed that the individual can live only up to 100 years for simplicity and practicality of the model).

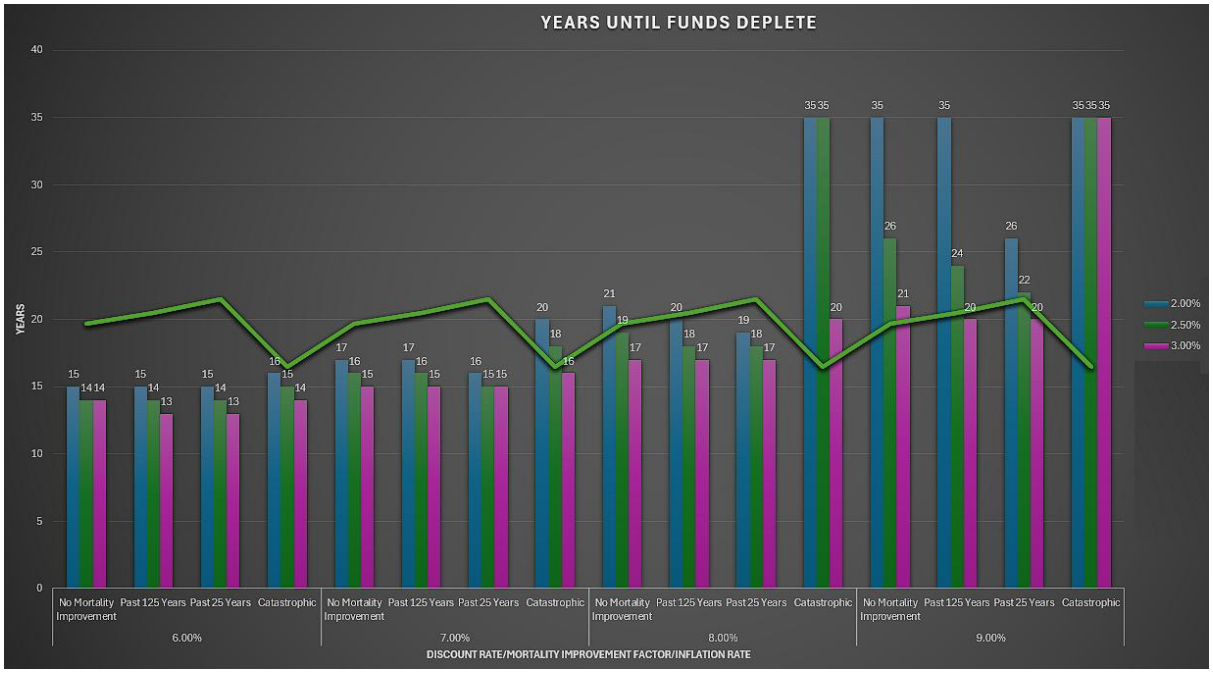

Figure 14: Years until funds are depleted for males

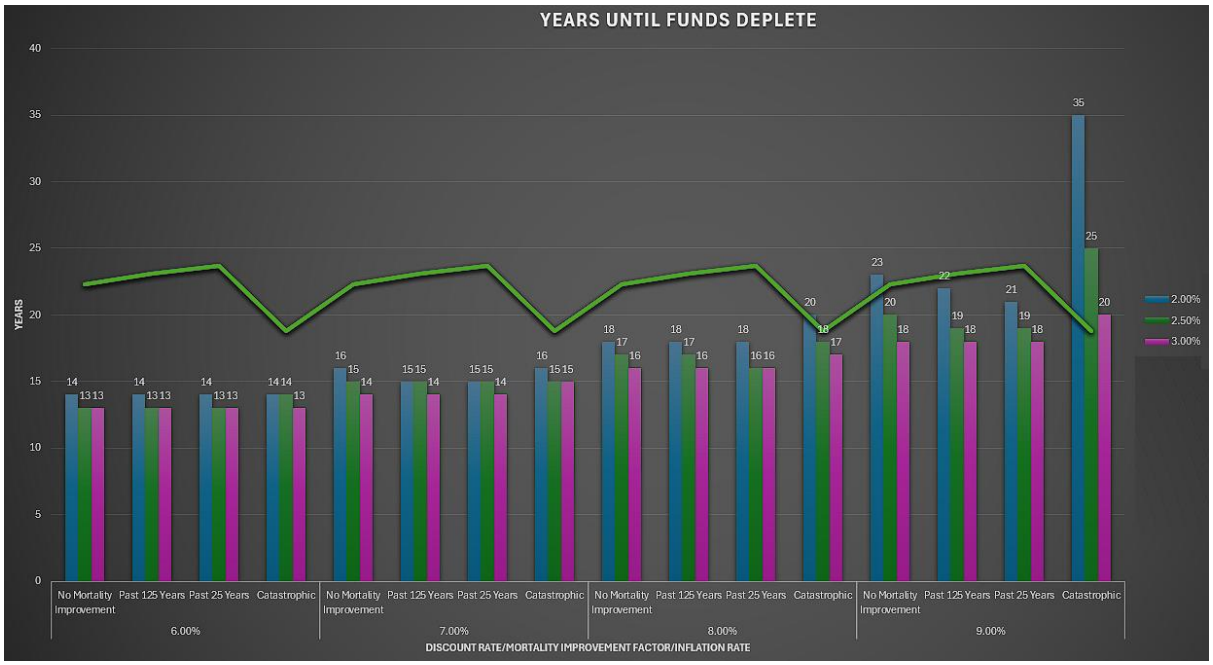

Figure 15: Years until funds are depleted for females

Notably, the funds the males have will more than exceed how long they will live in certain scenarios. As considering the case of the catastrophic scenario, at each catastrophic scenario the life expectancy is the lowest. The EPV for $100,000 growing at the rates of inflation, and for the selected mortality improvement type does not exceed $1,000,000, the amount of lump sum the individual has at the beginning of retirement. As the amount of funding required to support their desired retirement lifestyle is less than the funding they have on hand at the beginning of retirement, the funds that an individual has will still remain even though the individual will not.

In other parts of the graph (where the green line is above the bars), the EPV for $100,000 growing at the rates of inflation and select mortality improvements, exceeds the $1,000,000 the individual has available for retirement. In other words, the funding required for the desired income schedule exceeds what the individual has for retirement, and hence the individual is thus underfunded. The implication of this being that the individual will risk not having enough

funding for their desired lifestyle and hence outlive their funds unless they find another way to continue their income stream.

The reason why for females, the funds don't deplete at 35 years for the corresponding scenarios for males, is due to the expected accumulated savings for females being greater than males at any given age. This is due to females having a greater life expectancy, which leads to a greater weighting on the amount of expected accumulated savings required compared to males (who are more likely to die sooner before they require the funds at future ages). As the amount of expected accumulated savings exceeds the amount of lump sum they have on hand for retirement, being more than $1,000,000, the individual will deplete their funds sooner for their desired retirement income schedule.

For increases in discount rate, the funds take longer to deplete, as the increased investment rate of return provides more interest. And the increased interest will take longer to deplete than interest received from a lower rate of return. This is seen across corresponding mortality improvement types and inflation rate scenarios. In other words, for a catastrophic mortality improvement scenario and 2% inflation, the number of years until the funds deplete increases as the discount rate increases.

The mortality rate improvement types are ranked. With Catastrophic being the lowest improvement (more specifically, a deterioration), then No Mortality Improvement, then the past 125 Years of Mortality Improvement and finally the past 25 Years of Mortality Improvement. In many cases, improvements in mortality shows no significant increase in the number of years until the funds deplete except in the case of the larger discount rates in general. For increases in the mortality improvement, the number of years until the fund deplete seems to deplete sooner. An individual with a lower mortality rate is more likely to live longer and hence will be more likely to deplete their funds to sustain their longevity. Compared to an individual with a greater mortality rate, who is more likely to die sooner, and hence will have

on average, their funds will last longer than them assuming expenditure is the same. If an individual with a lower mortality rate does not lower their yearly income to sustain over their expected increased lifespan, then their funds will deplete sooner.

Increases in inflation rate will mean that more funds are required to accommodate the increasing cost of living. Due to this, in general as the inflation rate increases, the funds deplete in fewer years.

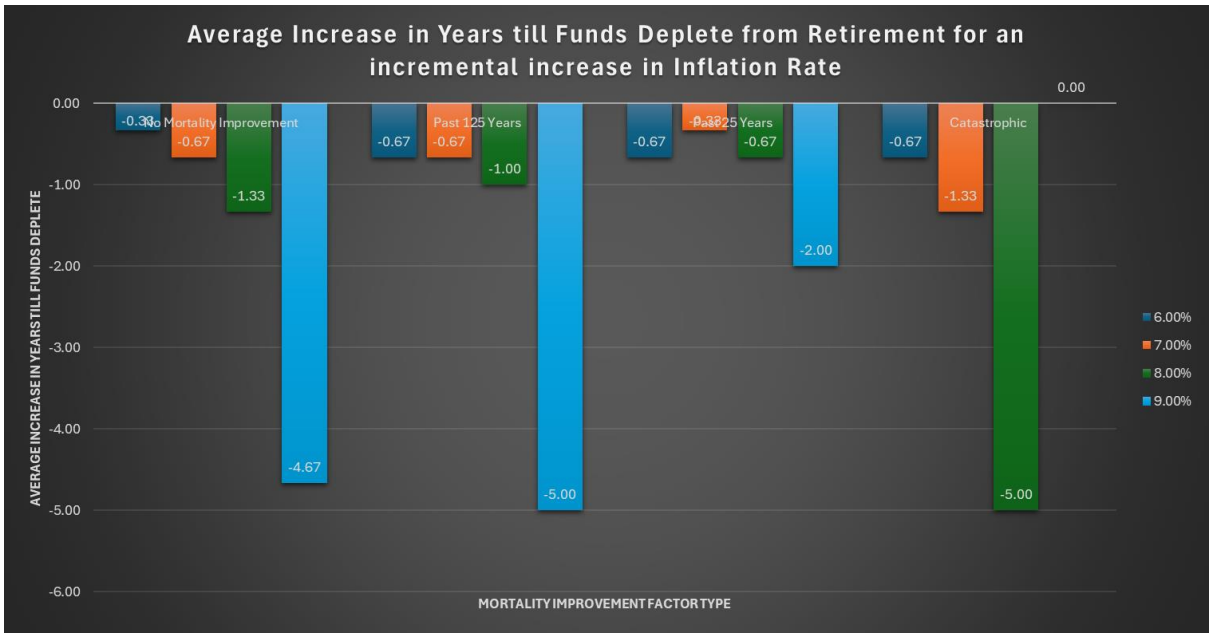

Figure 16: Average Increase in Years till Funds Deplete from Retirement for specified conditions and desired annual income schedule for an incremental increase in Inflation Rate for males

The graph above shows that for increments in the inflation rate of 0.5%, on average decrease the time until the funds deplete for varying mortality improvements types and discount rates for males by about 1.33 years. For females the decrease is 0.92 years. The graph for females is similar to the graph for the males with slightly different values and has hence been put in the appendix.

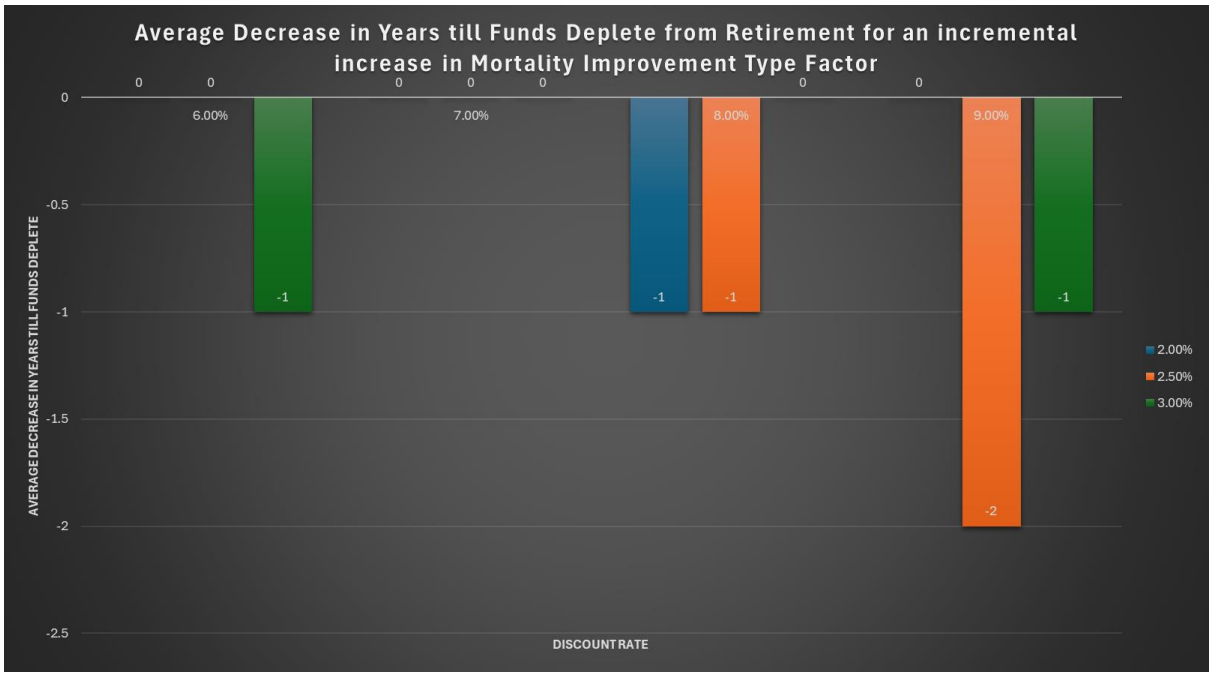

Figure 17: Average Decrease in Years till Funds Deplete from Retirement for specified conditions and desired annual income schedule for an incremental increase in Mortality Improvement Type Factor for males

The graph above shows that for increments in mortality improvement type, on average the funds deplete 1.58 years sooner for males for varying mortality improvements types and discount rates. For females this decrease is 0.58. The graph for females is similar to the graph for the males with slightly different values and has hence been put in the appendix.

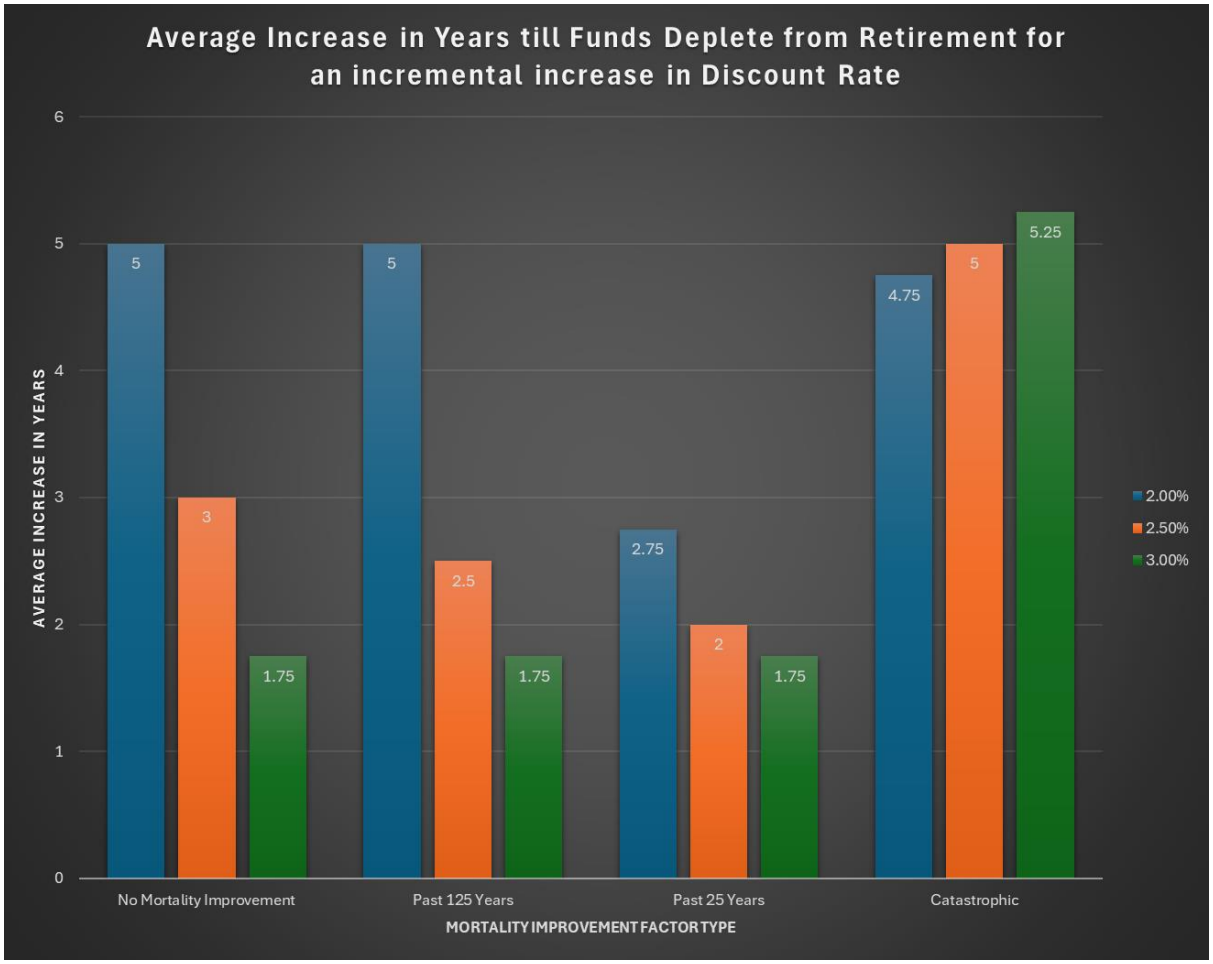

Figure 18: Average Increase in Years till Funds Deplete from Retirement for specified conditions and desired annual income schedule for an incremental increase in Discount Rate for males

The graph above shows that for 1% increments in discount rate, on average increase the years until the funds deplete by about 3.10 years for males for varying mortality improvements types and inflation rates. For females this is about a 1.95 year increase. The graph for females is similar to the graph for the males with slightly different values and has hence been put in the appendix.

Limitations of Research

- Life Table for 2024 not available so had to generate theoretical life table probabilities using improvement factors using most recent mortality data (2020-2022 Life Table as mentioned in Methodology).

- Further impact of COVID or future events similar to COVID is unknown on future life table probabilities.

Conclusion

Objective 1:

The conclusion of the analysis above in regards to EPV is that the amount of funds required for retirement increases with increases in mortality improvement, and inflation rate where as the funds required for retirement decreases with increases in discount rate.

Objective 2:

The conclusion of the analysis above in regards to the years until the lump sum of a retiring individual depletes is that the depletion rate increases with increased mortality improvement and inflation rate, whereas the funds depletion rate decreases with increases in discount rate.

The result of the research to solve Objectives 1 and 2, results in solving Objective 3 which are the following rules of thumbs which act more as guidelines or crude measures rather than actual results.

Rules of thumbs provide for a mental calculation of how EPV and the years until the funds deplete are affected to small incremental changes in assumptions. In this case, it is done within the 6%-9% parameters for discount rate, the mortality improvement types, inflation rates as shown in the table below:

Mortality Improvement Type	Discount Rate	Inflation Rate
No Mortality Improvement	6%	2%
Past 25 Years Improvement	7%	2.5%

Past 125 Years Improvement	8%	3%
Catastrophic	9%	

For every increment increase in mortality improvement type the expected accumulated savings required increases by around 0.8% for males and females and the number of years until the funds deplete decreases by 0.50 years for males and 0.25 years for females, assuming all other conditions stay constant.

For every 1% increment increase in discount rate the expected accumulated savings required decreases by around 6% for both males and females and the number of years until the funds deplete increases by 3.38 years for males and 2.02 years for females, assuming all other conditions regarding inflation and mortality improvement type stay constant.

Assuming cash flows increase at the rate of inflation, for every 0.5% increment in inflation rate, the expected accumulated savings required increases by 3% for both males and females, and the number of years until the funds deplete decreases by 1.5 years for males and 1 year for females, assuming all other conditions stay constant.

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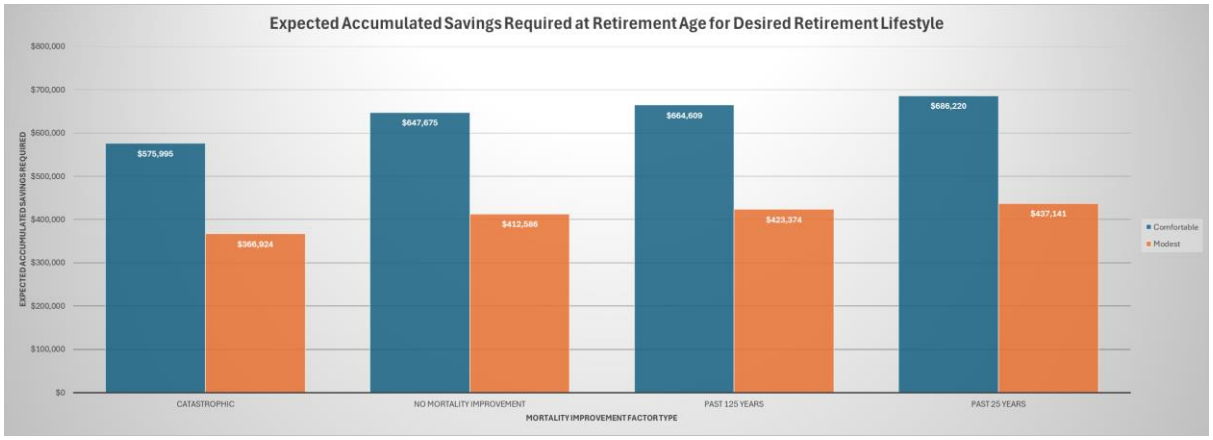
APPENDIX

Males:

Graphs Relating to Expected Accumulated Savings

A.



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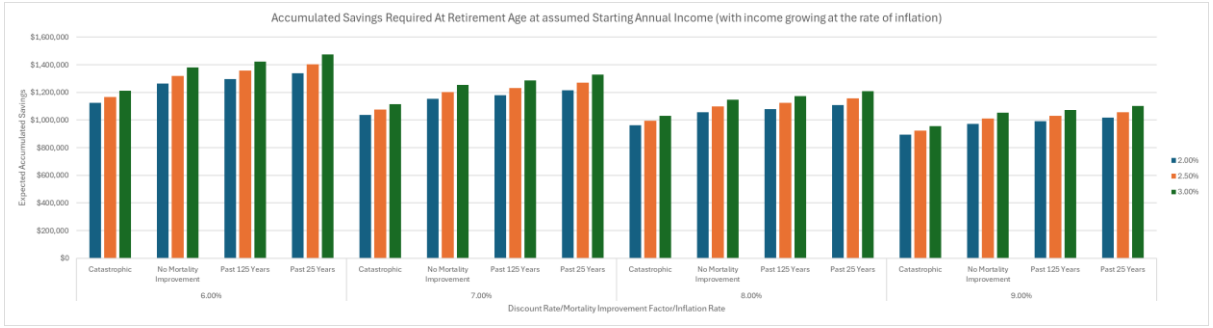

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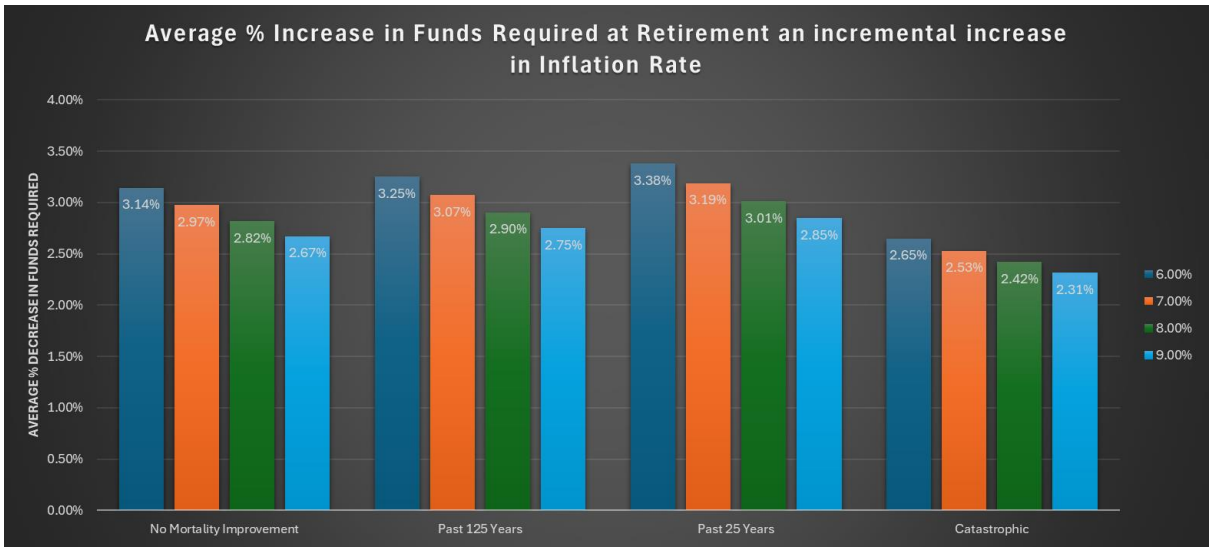

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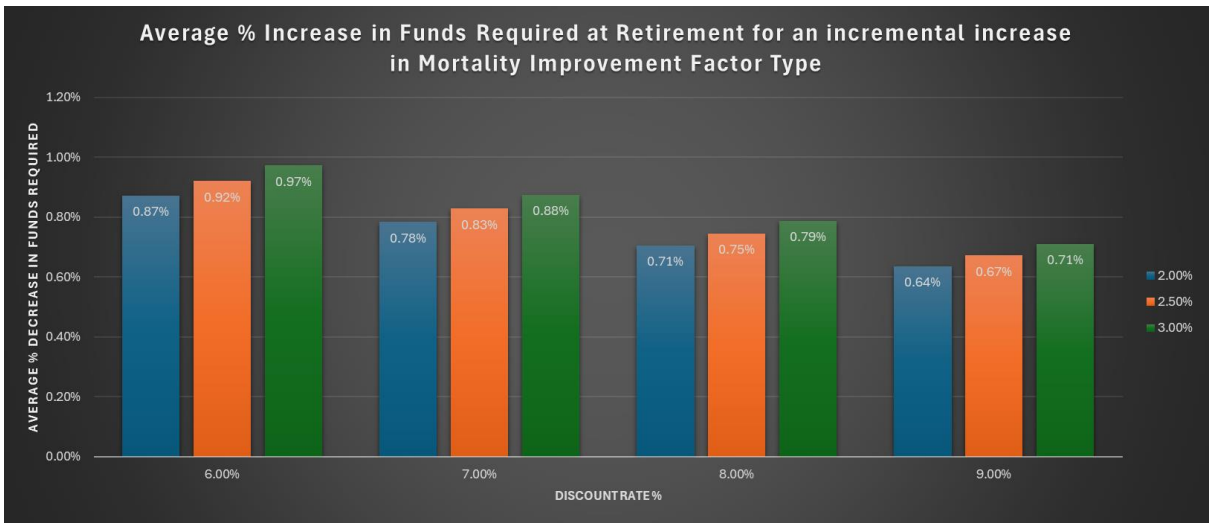

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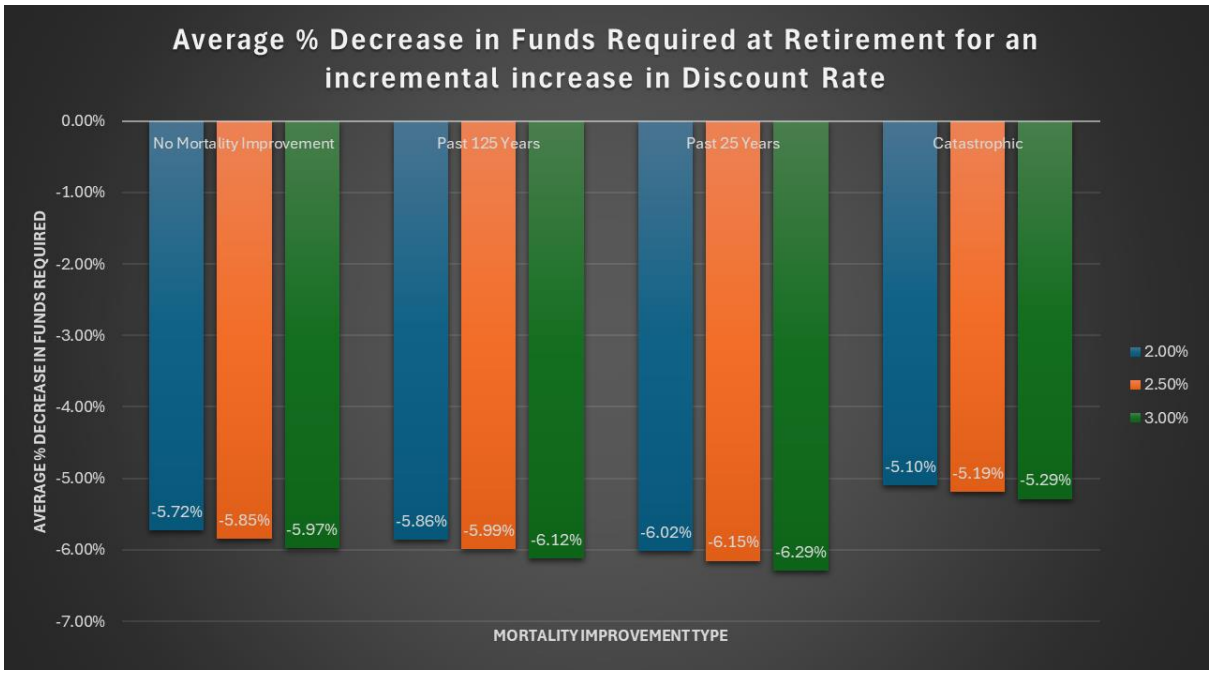
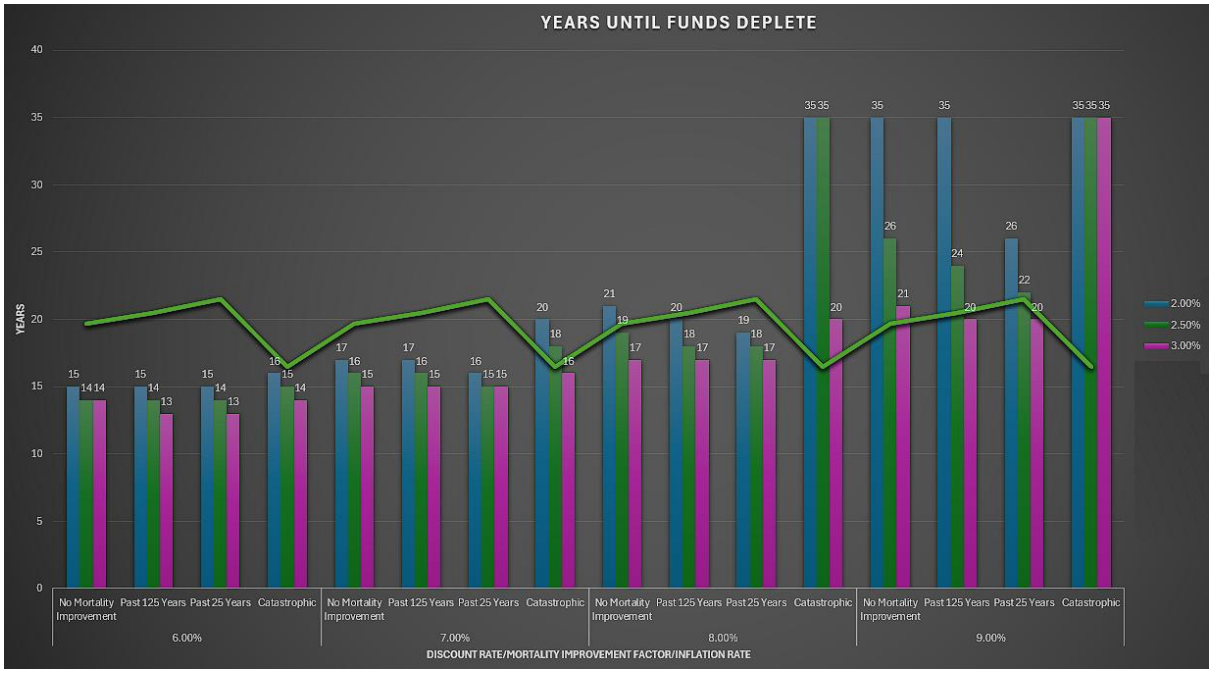
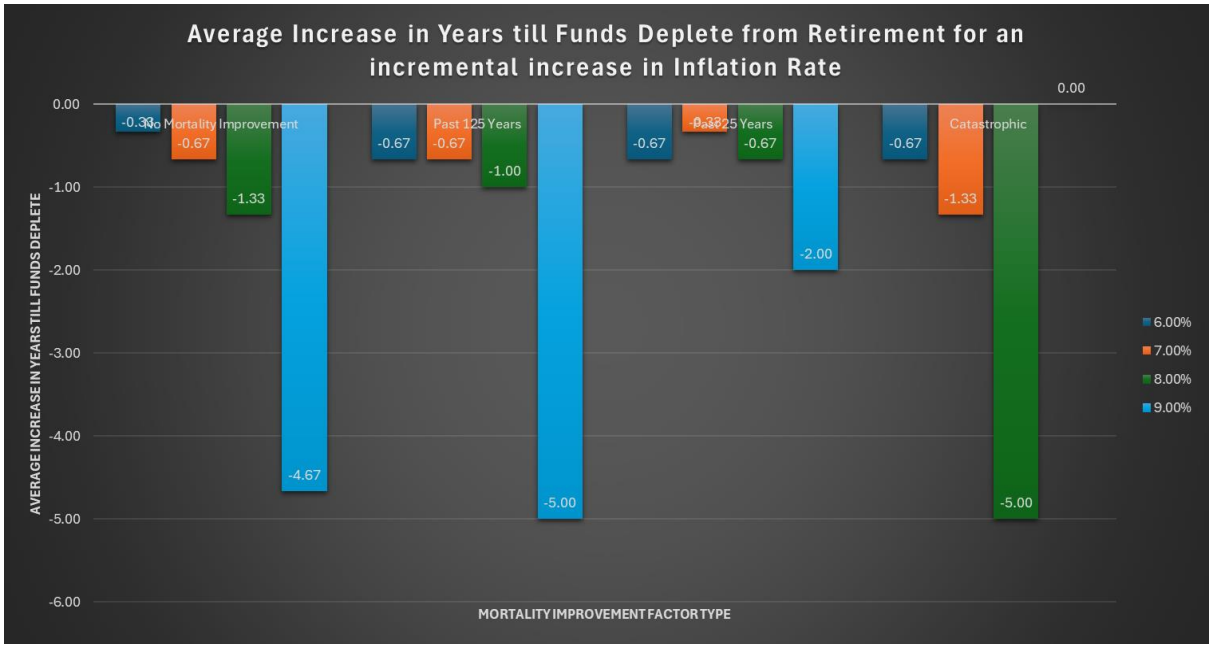


Graphs Relating to the Years Until Funds Deplete

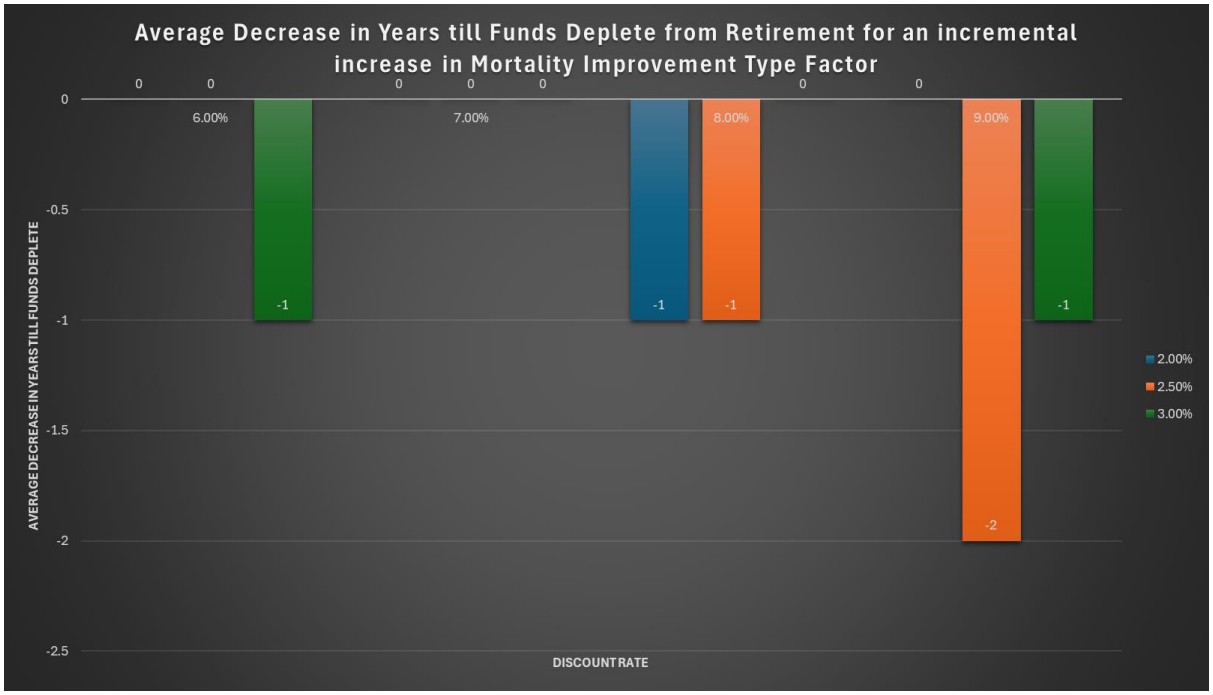
F.


YEARS UNTIL FUNDS DEPLETE
YEARS
No Mortality Improvement
Past 125 Years
Past 25 Years
Catastrophic
6.00%
7.00%
8.00%
9.00%
DISCOUNT RATE/MORTALITY IMPROVEMENT FACTOR/INFLATION RATE
2.00%
2.50%
3.00%

G.


Average Increase in Years till Funds Deplete from Retirement for an incremental increase in Inflation Rate
AVERAGE INCREASE IN YEARS TILL FUNDS DEPLETE
No Mortality Improvement
Past 125 Years
Past 25 Years
Catastrophic
MORTALITY IMPROVEMENT FACTOR TYPE
6.00%
7.00%
8.00%
9.00%

H.


Average Decrease in Years till Funds Deplete from Retirement for an incremental increase in Mortality Improvement Type Factor
AVERAGE DECREASE IN YEARS TILL FUNDS DEPLETE
DISCOUNT RATE
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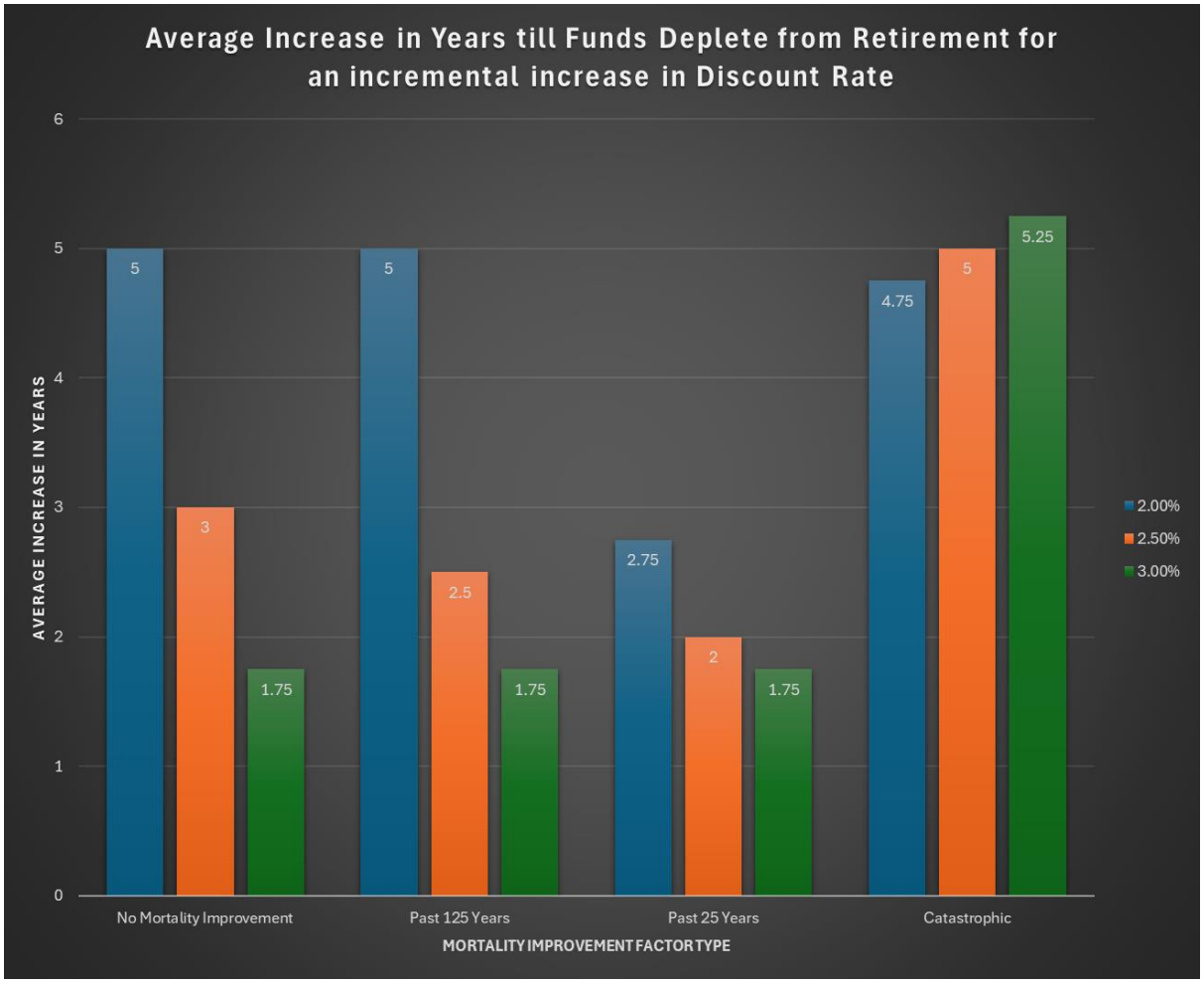
7.00%
8.00%
9.00%
2.00%
2.50%
3.00%

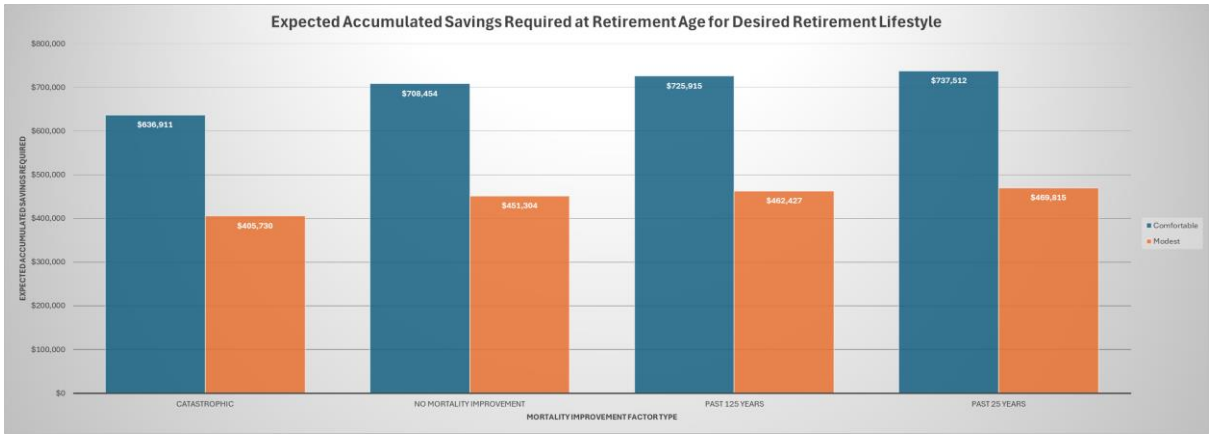
I.


Average Increase in Years till Funds Deplete from Retirement for an incremental increase in Discount Rate
AVERAGE INCREASE IN YEARS
MORTALITY IMPROVEMENT FACTOR TYPE
No Mortality Improvement
Past 125 Years
Past 25 Years
Catastrophic
2.00%
2.50%
3.00%

Females:

Graphs Relating to Expected Accumulated Savings

J.



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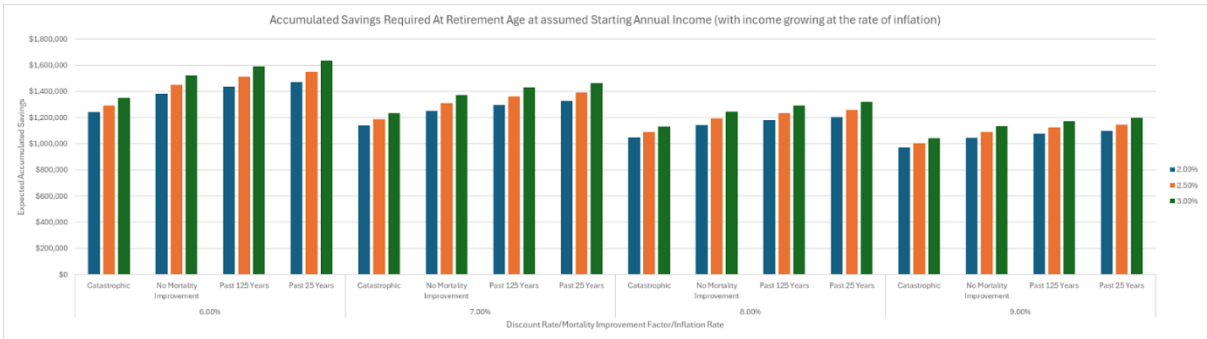

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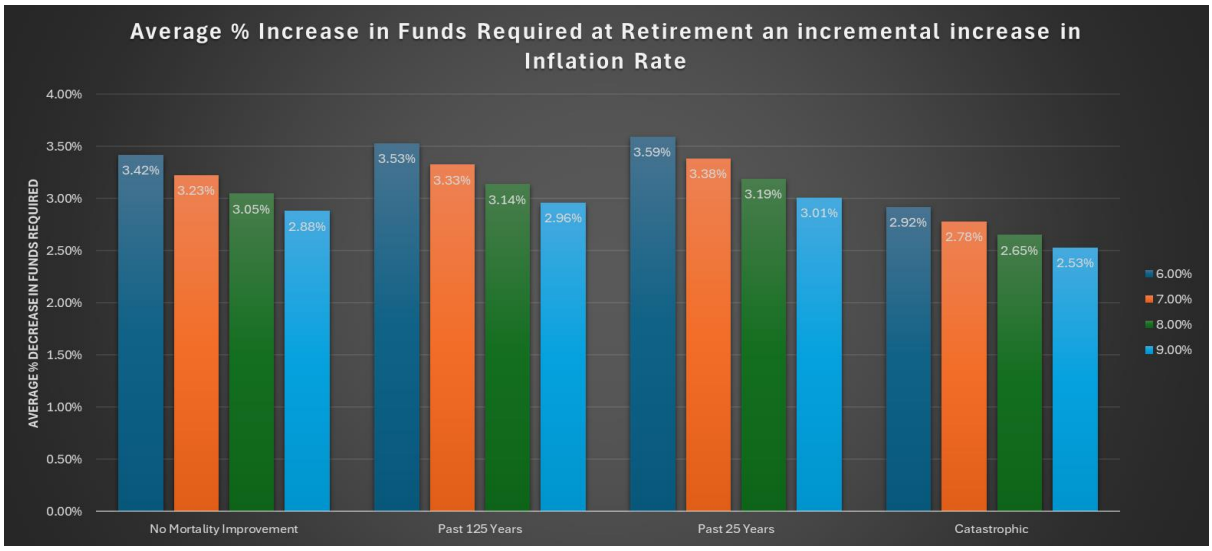

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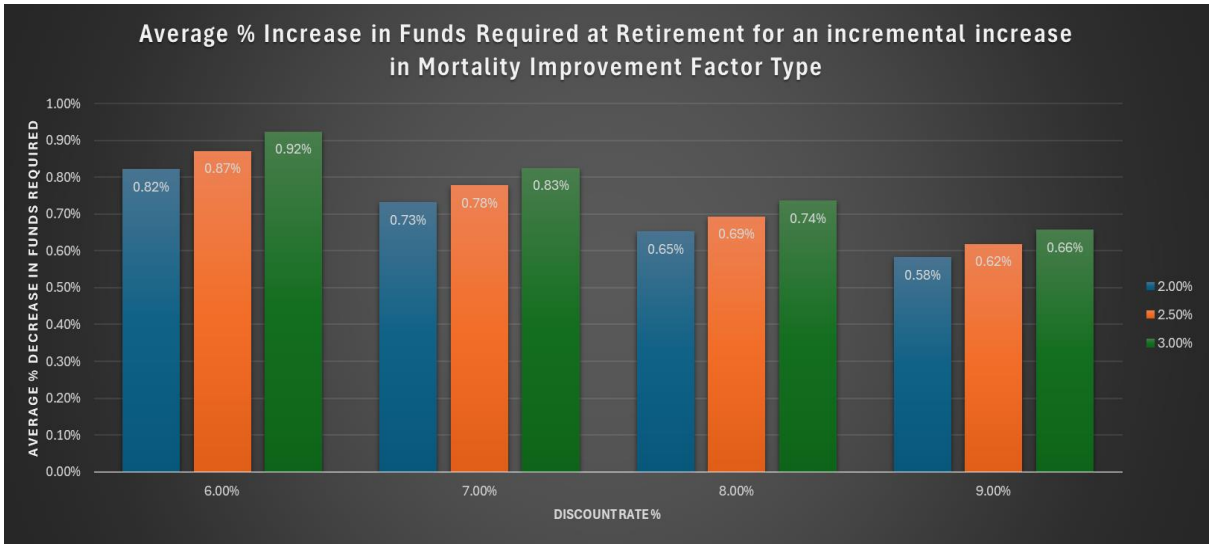

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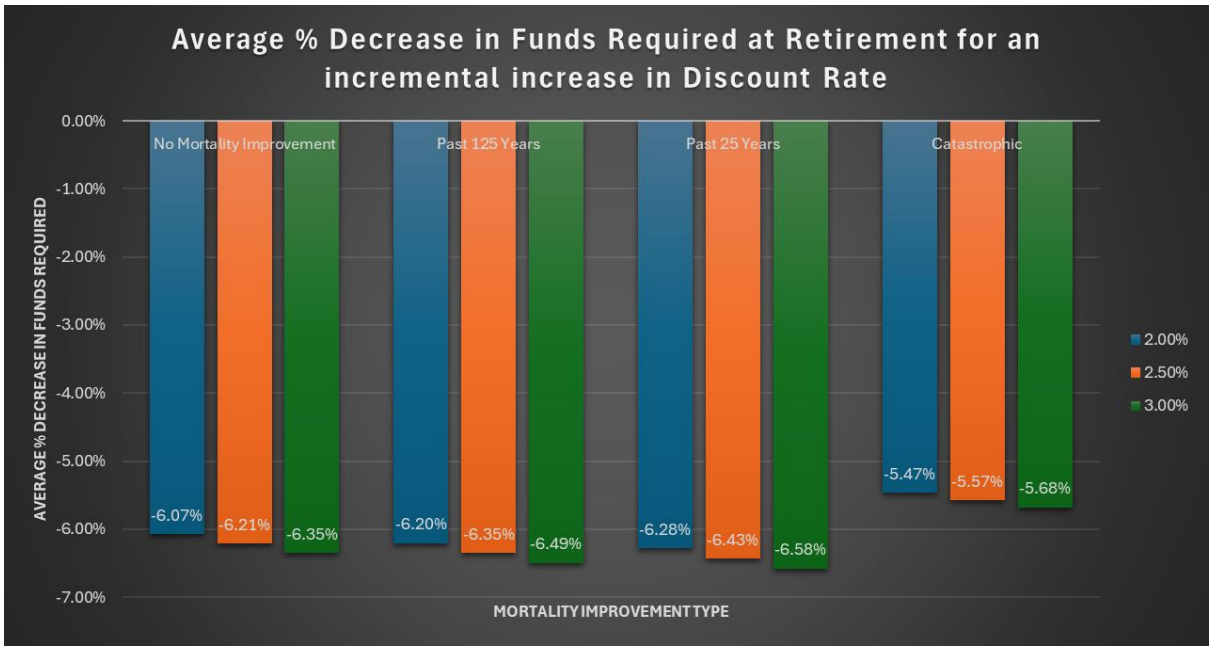

Graphs Relating to the Years Until Funds Deplete

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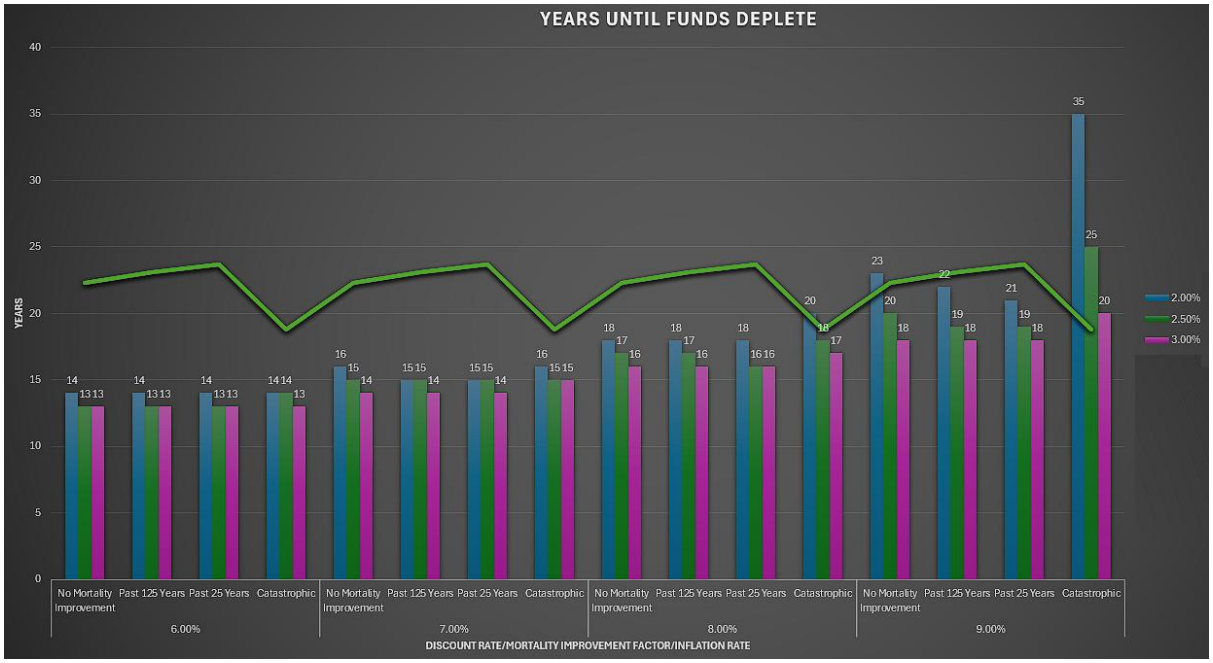
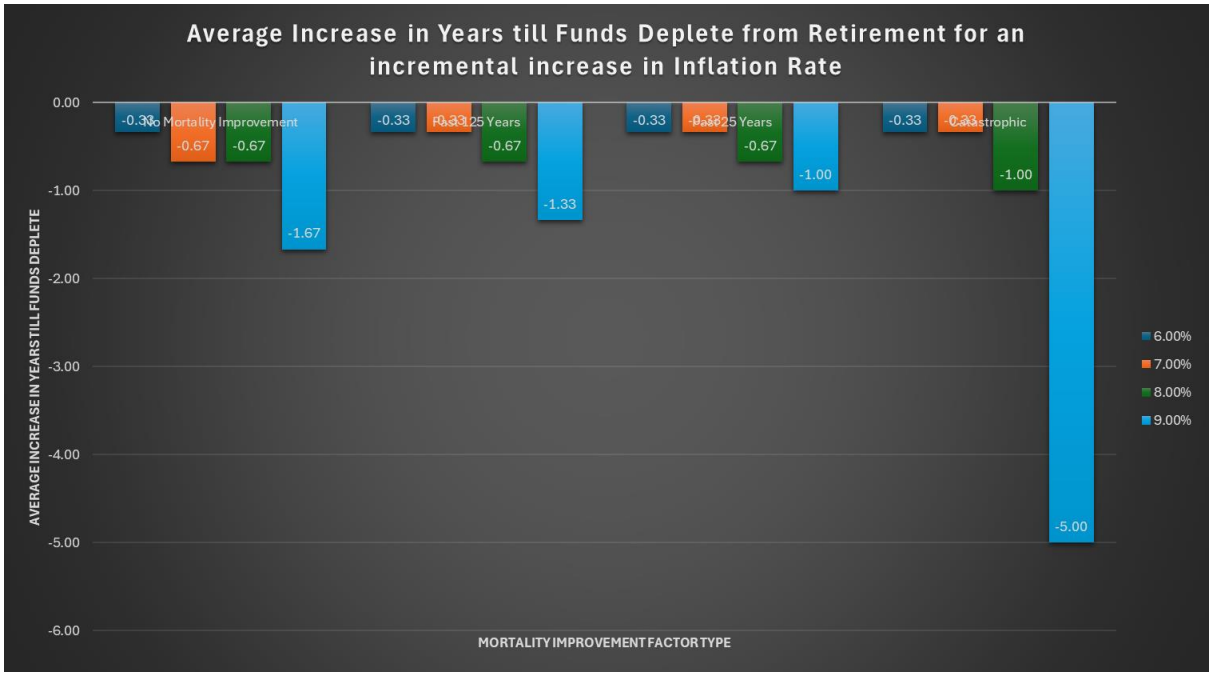
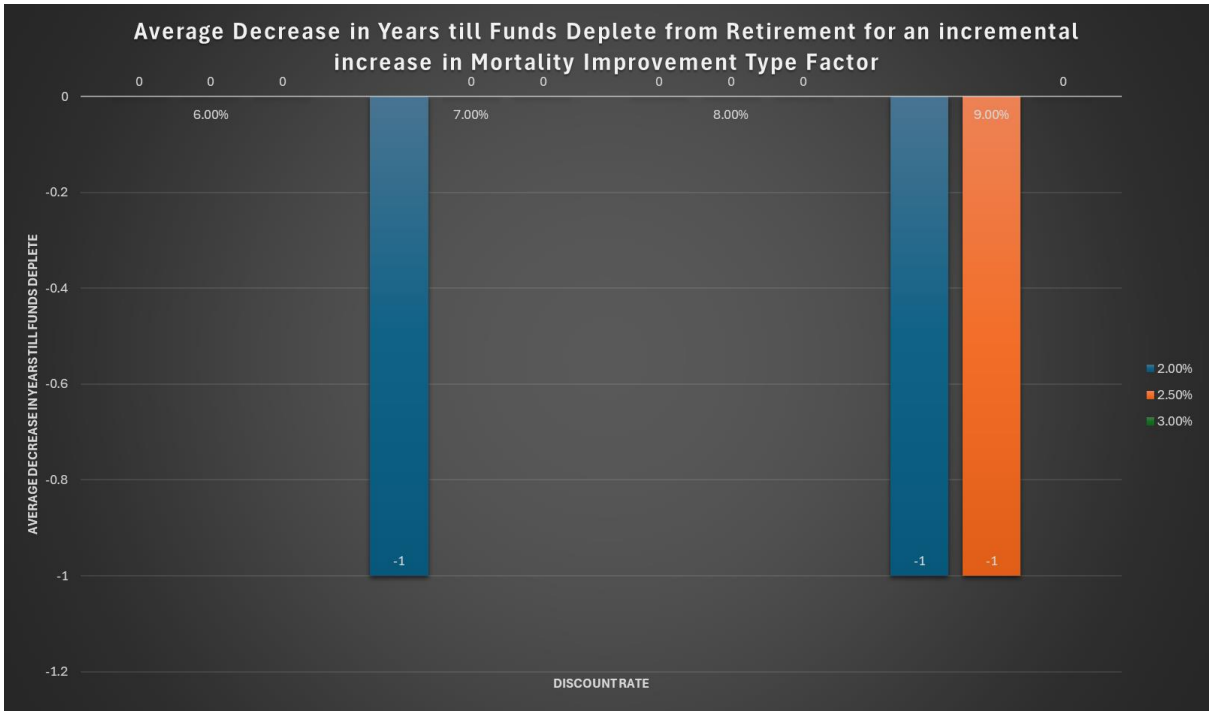

YEARS UNTIL FUNDS DEPLETE
YEARS
DISCOUNT RATE/MORTALITY IMPROVEMENT FACTOR/INFLATION RATE
No Mortality Improvement
Past 125 Years
Past 25 Years
Catastrophic
6.00%
7.00%
8.00%
9.00%
2.00%
2.50%
3.00%

P.


Average Increase in Years till Funds Deplete from Retirement for an incremental increase in Inflation Rate
AVERAGE INCREASE IN YEARS TILL FUNDS DEPLETE
MORTALITY IMPROVEMENT FACTOR TYPE
No Mortality Improvement
Past 125 Years
Past 25 Years
Catastrophic
6.00%
7.00%
8.00%
9.00%

Q.


Average Decrease in Years till Funds Deplete from Retirement for an incremental increase in Mortality Improvement Type Factor
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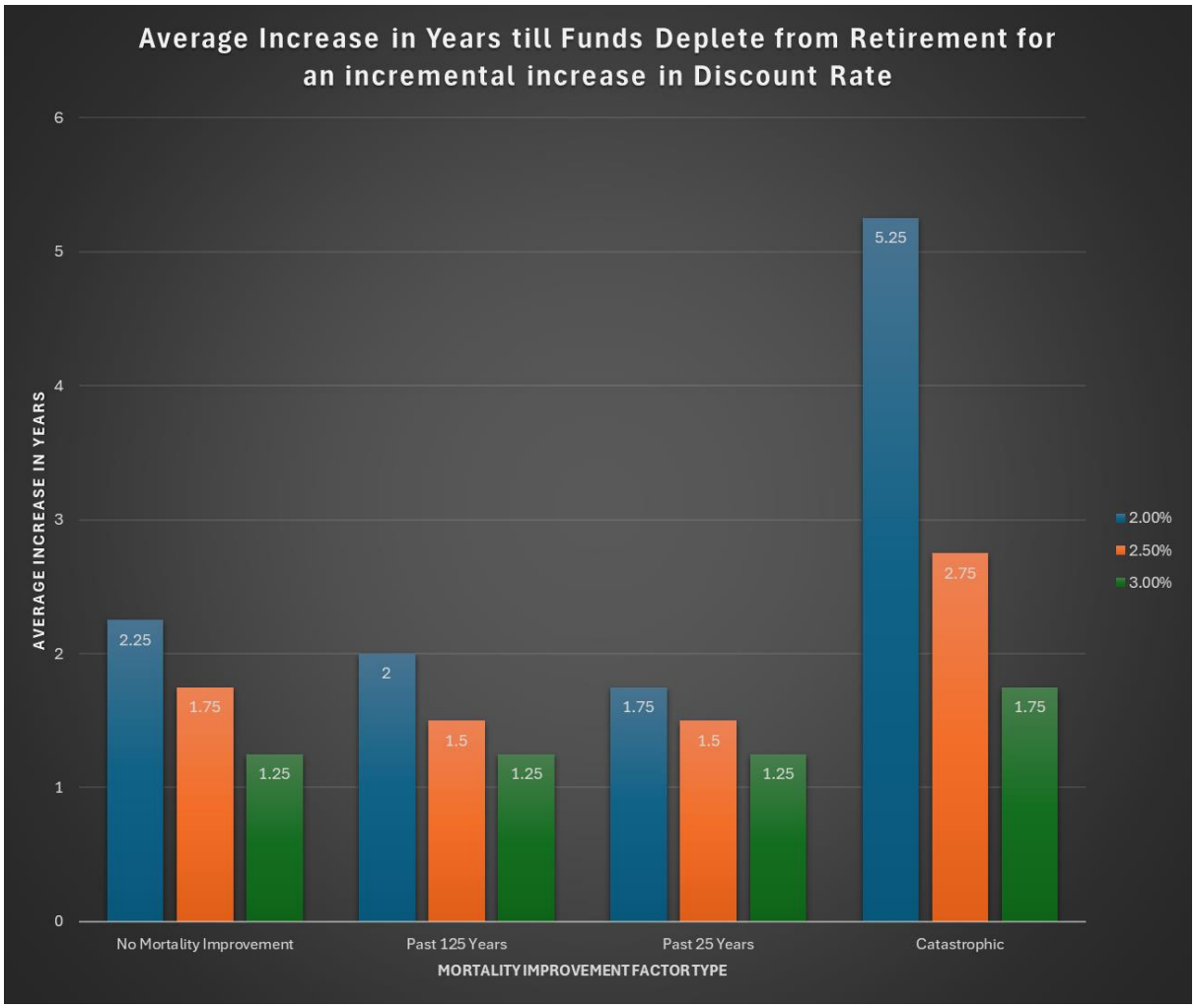
0
0
0
0
0
0
0
0
6.00%
7.00%
8.00%
9.00%
0
-0.2
-0.4
-0.6
-0.8
-1
-1.2
-1
-1
-1
AVERAGE DECREASE IN YEARS TILL FUNDS DEPLETE
DISCOUNT RATE
2.00%
2.50%
3.00%

R.


Average Increase in Years till Funds Deplete from Retirement for an incremental increase in Discount Rate
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5
4
3
2
1
0
2.25
1.75
1.25
2
1.5
1.25
1.75
1.5
1.25
5.25
2.75
1.75
AVERAGE INCREASE IN YEARS
No Mortality Improvement
Past 125 Years
Past 25 Years
Catastrophic
MORTALITY IMPROVEMENT FACTOR TYPE
2.00%
2.50%
3.00%